
THE SHAPE OF ATTRACTION IN UMAP: EXPLORING THE EMBEDDING FORCES IN DIMENSIONALITY REDUCTION

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ABSTRACT

Uniform manifold approximation and projection (UMAP) has established itself as one of the pre-eminent neighbor embedding methods, procuring its low-dimensional representations through the interplay of attractive and repulsive forces exerted among high-dimensional data points. In the present work, we undertake a systematic dissection of these forces so as to expose the mechanisms by which they govern cluster formation and visualization, and we situate UMAP relative to its contemporaries through a comparative analysis. Whereas repulsion accentuates dissimilarity—thereby delineating cluster boundaries and regulating inter-cluster separation—attraction proves considerably more elusive, since the attractive tension subsisting between a pair of points may, in the lower-dimensional mapping, manifest simultaneously as both attraction and repulsion. This duality accounts for the indispensability of learning-rate annealing and furnishes a principled rationale for the divergent treatment accorded to the attractive and repulsive terms. Furthermore, through a targeted modification of attraction, we demonstrably enhance the consistency with which clusters form under random initialization. Taken as a whole, our analysis renders UMAP and cognate embedding methods appreciably more interpretable, more robust, and more accurate.

1 Introduction

The contemporary scientific landscape is defined by an ever-accelerating proliferation of high-dimensional data. Dimensionality reduction (DR) techniques have consequently emerged as indispensable instruments for the exploratory analysis of such data, affording a visualization of the structure latent within it. The most widely adopted among these methods— t -distributed stochastic neighbor embedding (Maaten and Hinton, 2008) and uniform manifold approximation and projection (UMAP) (McInnes et al., 2018)—are alike predicated upon attraction–repulsion dynamics, whereby similar data points are drawn together while dissimilar ones are driven apart. Being unsupervised in nature, these algorithms dispense entirely with labeled data; rather, they discern and preserve the intrinsic structure of high-dimensional data by exploiting local (attractive) and global (repulsive) relationships, or forces. This attribute renders them especially well suited to a diverse array of tasks—among them clustering (Becht et al., 2019), exploratory data analysis (Fleischer and Islam, 2020), anomaly detection in semiconductor manufacturing (Fan et al., 2021), visual search (González-Márquez et al., 2024), time series analysis (Altin and Cakir, 2024), the study of representation convergence (Huh et al., 2024), and outlier image detection (Islam and Fleischer, 2024)—in each of which the visualization of concealed patterns within unlabeled data proves both critical and consequential. By inferring embeddings in a data-driven, label-free manner, DR epitomizes the capacity of unsupervised methods to distill intricate data into readily interpretable forms.

Elaborating upon this attraction–repulsion principle, a succession of newer methods has since arisen (Amid and Warmuth, 2019; Agrawal et al., 2021; Wang et al., 2021; Narayan et al., 2021; Yang et al., 2022; Wang et al., 2025; Kury et al., 2025), each contrived to foreground particular facets of the data. Notwithstanding their pertinence across a broad spectrum of applications, these methods frequently repose upon heuristic expedients that may fail to yield meaningful interpretations. DR, moreover, introduces distortions that are constitutively unavoidable (Chari and Pachter, 2023). It is therefore imperative to cultivate a more penetrating understanding of the underlying algorithms, so that practitioners might interpret the resulting embeddings more faithfully, forestall the emergence of spurious structures, and optimize performance. In practice, these algorithms attain compact clusters by recourse to a variety of devices—among them bespoke initialization, learning-rate scheduling, and kernel-function tuning. Yet the dynamics of the attractive and repulsive forces that underlie cluster formation have never been subjected to thorough investigation. Compounding

the difficulty, the essential tunable parameters lie concealed within abstract functional forms, thereby impeding any transparent explanation of the algorithms.

In the present paper, we decompose these forces into their constituent parts and thereby extract the functional shapes of attraction and repulsion. We establish that the necessity of learning-rate annealing, the difficulty of securing consistent output under random initialization, and the very genesis of cluster formation are all traceable to attraction. Repulsive forces, by contrast, principally govern the distances obtaining between clusters. Our specific contributions are the following:

1. We derive the attraction and repulsion shapes from the attractive and repulsive forces, establish the conditions under which distance contracts and expands, and thereby afford a fresh vantage point upon these algorithms (Section 4).
2. We demonstrate that the attraction shape of UMAP engenders the counterintuitive coexistence of both contraction and expansion of distance; and, by juxtaposing the attraction shapes of distinct algorithms, we elucidate the manner in which attraction conditions the learning-rate annealing scheme (Section 5).
3. We modify the attraction shape so as to enhance the consistency of the embedding under random initialization—an outcome indicative of the encoding of a unique underlying structure (Section 5.1).
4. Through an analysis of the repulsion shapes, we furnish a deeper understanding of cluster formation and of the regulation of inter-cluster distance (Section 5.2).
5. We compare the attraction and repulsion shapes of UMAP, Parametric UMAP, and NEG- t -SNE, disclose the affinities and distinctions among them, and characterize the stability of the respective algorithms (Section 6).

We center the main text upon UMAP—now a de facto standard across numerous fields—and adduce other algorithms wherever germane. For a fuller treatment of the several algorithms, the reader is referred to the appendix. The code employed in this research will be released upon acceptance.

2 Related Works

The provenance of modern iterative graph-based neighbor embedding algorithms may be traced to stochastic neighbor embedding (SNE) (Hinton and Roweis, 2002) and to its extension by way of the t -distribution (t -SNE) (Maaten and Hinton, 2008). Both methods rest upon a dense graph, wherein every point in a dataset entertains a pairwise relation with all the others, irrespective of whether the points concerned are mutually similar; the weights of these graphs are, moreover, normalized so as to confer upon them the character of a probability distribution. Other concurrent methods—locally linear embedding (Roweis and Saul, 2000) and Laplacian Eigenmaps (LE) (Belkin and Niyogi, 2002) among them—instead employed a k -nearest neighbor (k -NN) graph of pairwise interactions; designated spectral methods, these algorithms are predicated upon eigenvalue decomposition. Subsequent work by Tang et al. (Tang et al., 2016) assimilated the k -NN graph into the iterative framework and dispensed with normalization in the lower dimension. This approach was carried further by McInnes et al. (2018) in UMAP, wherein the normalization step was abolished altogether—in both the high and low dimensions alike—and the embedding was procured through pairwise interactions alone. Its optimization steps invoke an explicit attractive force to preserve the local neighborhood alongside a repulsive force to hold dissimilar points apart. Building upon these foundations, methods such as PaCMAP (Wang et al., 2021) and NEG- t -SNE (Damrich et al., 2023) have subsequently been advanced. For a recent survey of the field, the reader is directed to de Bodt et al. (2025).

Considerable progress has been achieved in apprehending and explicating the relationships that obtain among these algorithms. An early analysis of SNE established that data well clustered in the original space remain well clustered in the embedding space (Shaham and Steinerberger, 2017). A cognate analysis of t -SNE by Linderman and Steinerberger (2019) demonstrated that the number of clusters manifest in the embedding space constitutes a lower bound on the number present in the original space—a result subsequently refined through further characterization (Arora et al., 2018; Cai and Ma, 2022; Linderman and Steinerberger, 2022). Inasmuch as t -SNE and UMAP issue from a common underlying framework yet yield drastically divergent visualizations, a principal preoccupation of the literature has been the elucidation of the connection between them (Böhm et al., 2022; Damrich et al., 2023; Draganov et al., 2023). Böhm et al. (2022) advanced the thesis that methods such as Laplacian Eigenmap, ForceAtlas2 (Jacomy et al., 2014), UMAP, and t -SNE all constitute samples drawn from one and the same underlying spectrum. Indeed, LE and t -SNE are bound together through the early-exaggeration phase (Cai and Ma, 2022), while the affinity between UMAP and t -SNE may be articulated by way of contrastive estimation (Damrich et al., 2023). Contemporaneously, Hu et al. (2023) independently uncovered the relation between contrastive learning and SNE. More recent contributions bring a probabilistic perspective to bear (Ravuri et al., 2023; Ravuri and Lawrence, 2024), deploy kernel techniques (Draganov

and Dohn, 2023), and enlist information geometry (Kolpakov and Rocke, 2024) in the effort to explain dimensionality reduction.

3 Uniform Manifold Approximation and Projection

UMAP constructs a high-dimensional graph of the dataset $X = \{x_i \in \mathbb{R}^n | i = 1, \dots, N\}$ by recourse to a pairwise relation, $p_{i,j} = f_h(d(x_i, x_j))$, which typically resides $\in [0, 1]$ (for the particulars, see Appendix H).

The graph of the low-dimensional representation $Y = \{y_i \in \mathbb{R}^d | i = 1, \dots, N\}$ is, in turn, prescribed by a differentiable function

$$q_{ij} = \frac{1}{1 + a(\|y_i - y_j\|_2^2)^b}, \quad (1)$$

where the parameters a and b determine the density of the mapping and are chosen by fitting q_{ij} to

$$\Psi(d_{ij}) = \begin{cases} 1 & \text{if } \|y_i - y_j\|_2 < m_d, \\ \exp(-(d_{ij} - m_d)) & \text{otherwise} \end{cases}, \quad (2)$$

wherein m_d regulates the distance separating the two nearest low-dimensional points. UMAP seeks to minimize the following cross-entropy loss function:

$$\mathcal{L} = \sum_{i,j} (-p_{ij} \log(q_{ij}) - (1 - p_{ij}) \log(1 - q_{ij})). \quad (3)$$

The first term furnishes an attractive force, whereas the second furnishes a repulsive one. Rather than optimizing every point at each iteration, UMAP adopts the negative-sampling stratagem (Mikolov et al., 2013; Tang et al., 2016): for each edge with $p_{ij} > 0$ —termed a positive edge—several further edges, termed negative edges, are drawn at random. The attractive force is brought to bear upon the positive edge:

$$y_i^{t+1} = y_i^t + \lambda \nabla_{y_i^t} \log(q_{ij}), \quad (4)$$

$$y_j^{t+1} = y_j^t + \lambda \nabla_{y_j^t} \log(q_{ij}), \quad (5)$$

and the repulsive force upon the negative edges:

$$y_i^{t+1} = y_i^t + \lambda \nabla_{y_i^t} \log(1 - q_{ij}), \quad (6)$$

where $\lambda(> 0)$ denotes the learning rate and t the step number. It bears noting that y_j is left unaltered for negative edges. For a detailed analysis of UMAP’s loss, the reader is referred to (Damrich and Hamprecht, 2021).

4 Attraction and Repulsion Shapes

The action of the updates (4, 6) admits of a considerable simplification once the gradients $\nabla_{y_i^t} \log(q_{ij})$ (and $\nabla_{y_i^t} \log(1 - q_{ij})$) are decomposed into a scalar coefficient—contingent upon the distance $\|y_i - y_j\|_2$ —acting upon the vector $(y_i - y_j)$. This scalar coefficient we designate the attraction (respectively, repulsion) shape. Although we adduce UMAP as a concrete exemplar, the formalism extends to any method predicated upon attraction and repulsion.

Writing $\nabla_{y_i^t} \log(q_{ij}) = f_a(\zeta^t)(y_i^t - y_j^t)$, with $\zeta = \|y_i - y_j\|_2$, we may recast the update equations of a positive edge as

$$y_i^{t+1} = y_i^t + \lambda f_a(\zeta^t)(y_i^t - y_j^t), \quad (7)$$

$$y_j^{t+1} = y_j^t - \lambda f_a(\zeta^t)(y_i^t - y_j^t). \quad (8)$$

Here $f_a : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\leq 0}$ constitutes the attraction shape, and we avail ourselves of the fact that, under the Euclidean metric, $\nabla_{y_i^t} \log(q_{ij}) = -\nabla_{y_j^t} \log(q_{ij})$. In analogous fashion, we reformulate the update equation of a negative edge as

$$y_i^{t+1} = y_i^t + \lambda f_r(\zeta^t)(y_i^t - y_j^t), \quad (9)$$

$$y_j^{t+1} = y_j^t, \quad (10)$$

where $f_r : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ constitutes the repulsion shape.

A decomposition of this kind has surfaced previously—for instance in (McInnes et al., 2018; Agrawal et al., 2021; Draganov and Dohn, 2023)—though the several formulations and their attendant uses diverge appreciably. The original UMAP paper (McInnes et al., 2018) enlisted it as a computational expedient in the service of rapid processing, whereas (Draganov and Dohn, 2023) pressed it into service for comparing distinct algorithms from a kernel-theoretic standpoint; in both instances the paramount concern was the computation of the derivative. Here, by contrast, we treat the decompositions as independent functions susceptible of assuming a variety of forms. A kindred approach was adopted earlier by (Agrawal et al., 2021), who nonetheless expressed the decomposition in terms of a scalar coefficient together with a unit vector $((y_i - y_j)/\|y_i - y_j\|_2)$ so as to accentuate the magnitude of the forces. The discussion that ensues will make plain that our shape decomposition proves more illuminating than magnitude considered in isolation.

4.1 Conditions for Attraction and Repulsion

In this section, we derive the conditions governing attraction and repulsion directly from the update equations (7)–(10). The proposition that follows characterizes the contraction of distance between the pair y_i^t and y_j^t constituting a positive edge:

Proposition 4.1. *The update Eqs. (7) and (8) provide a contraction of distance ($\|y_i^{t+1} - y_j^{t+1}\| < \|y_i^t - y_j^t\|$) if $-1 < \lambda f_a < 0$.*

Here λf_a functions as the effective attraction shape; in the preponderance of cases, however, f_a alone suffices to warrant meaningful conclusions concerning the embeddings. For a negative edge, the ensuing proposition characterizes the expansion of the distance between the pair y_i^t and y_j^t :

Proposition 4.2. *The update Eqs. (9) and (10) provide an expansion of distance ($\|y_i^{t+1} - y_j^{t+1}\| > \|y_i^t - y_j^t\|$) if $f_r > 0$.*

It bears remarking that the inclusion of a symmetric term $-\lambda f_r(\zeta)(y_i^t - y_j^t)$ in Eq. (10) in no way perturbs the conclusion set forth in this proposition. These conditions, moreover, furnish a per-iteration certificate of guaranteed contraction or expansion, and ought not to be conflated with the learning-rate tuning or decay mechanism, inasmuch as the shapes may be so devised that contraction and expansion do not hinge upon learning-rate decay whatsoever.

Figure 1(a) illustrates the effect exerted upon two points by differing values of $f_a < 0$ and $f_r > 0$. The latter shape is comparatively straightforward, since any positive value augments the distance. The former proves far more subtle, inasmuch as f_a encodes both attractive and repulsive dynamics: for $f_a \in (-1.0, 0)$ the distance contracts, with a sign flip occurring at the value $f_a = -0.5$ of maximal attraction (coincident points). Any value below -1.0 , by contrast, induces the distance to expand. Though these forces operate locally, they conspire collectively to shape the global structure.

5 Analysis of UMAP in terms of Attraction and Repulsion Shape

Using the gradient decomposition and the distance form (1), the attraction and repulsion shapes are given by

$$f_a^U(\zeta) = -\frac{2ab\zeta^{2(b-1)}}{1 + a\zeta^{2b}} \quad (11)$$

and

$$f_r^U(\zeta) = \frac{2b}{\zeta^{2b}(1 + a\zeta^{2b})}, \quad (12)$$

respectively. The present section concentrates upon the default shapes of UMAP—governed principally by the parameters a and b —and expounds the insights garnered by perturbing them. The discussion that follows holds in general for $b \leq 1$, the regime in which the attraction shape is strictly increasing and therefore invertible (for the derivation, see Appendix A).

Figure 1(b) displays the default attraction shape of UMAP ($a = 1.58, b = 0.89$), which grows unbounded (approaching ∞) as $\zeta \rightarrow 0$. As foretold by Proposition 4.1 the transition from contraction to expansion supervenes when λf_a^U crosses -1 as ζ approaches 0. Since the attraction shape is invertible, we may identify the distance at which this transition occurs, $\zeta_{-1} = (\lambda f_a^U)^{-1}(-1)$, as the minimum distance for contraction attributable to attractive updates. In effect, $\zeta > \zeta_{-1}$ engenders contraction whereas $0 < \zeta < \zeta_{-1}$ engenders expansion—a state of affairs that contravenes the intuition according to which attractive updates should invariably draw points nearer to one another.

Should ζ_{-1} be large, neighboring points oscillate between contraction and expansion and the clusters accordingly present a diffuse, fuzzy aspect. In pursuit of the sharpest boundaries, then, the objective of optimization may be recast

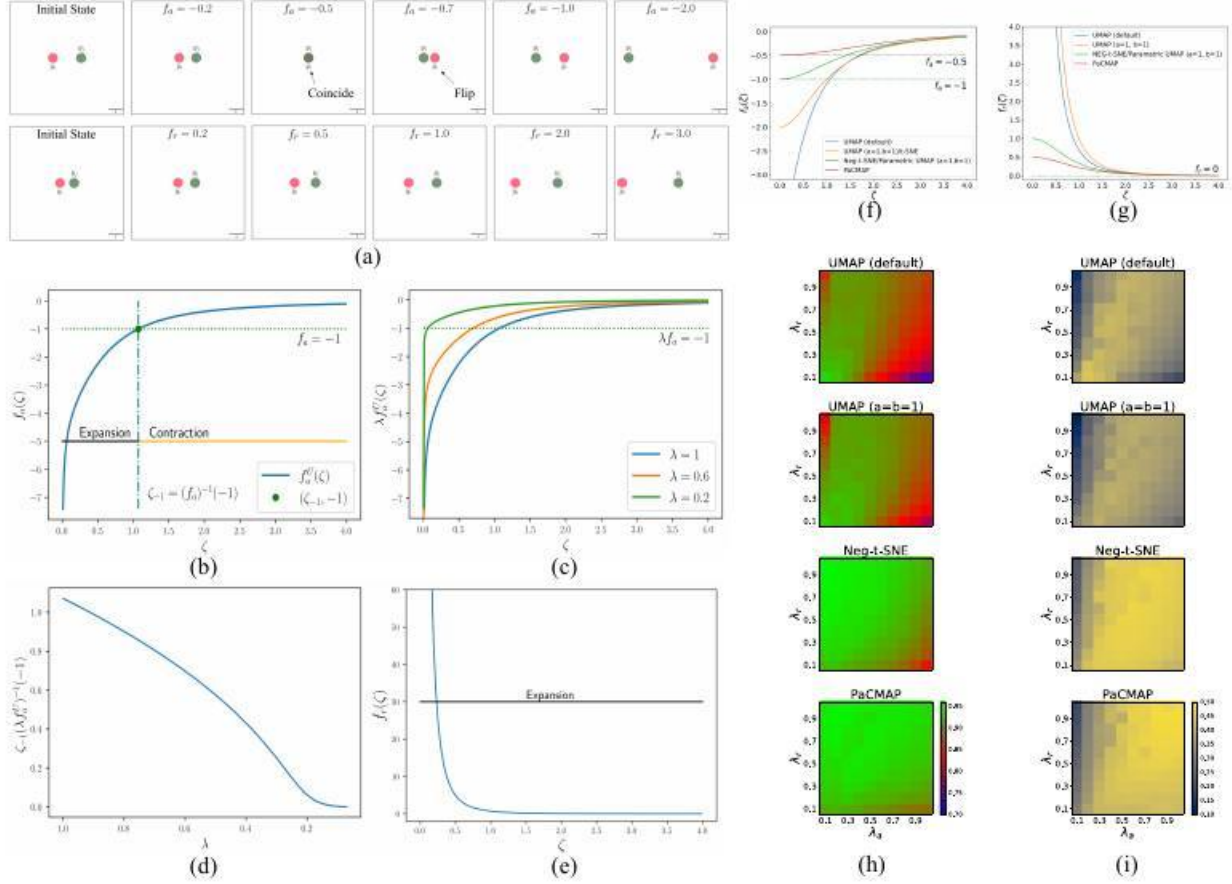


Figure 1: Attraction and repulsion shapes in UMAP. (a) The effect exerted upon a pair by differing values of f_a (top) and f_r (bottom). (b) The attraction shape of UMAP. (c) The effective attraction shape (λf_a) across various learning rates λ . (d) The minimum distance for contraction (ζ_{-1}) as λ diminishes. (e) The repulsion shape of UMAP. (f) Attraction and (g) repulsion shapes for a range of embedding methods. (h) Trustworthiness and (i) Silhouette Score of the various methods as the constant learning rate is varied independently for attraction (λ_a) and repulsion (λ_r). Default UMAP parameters: $a = 1.58$ and $b = 0.89$.

as the attainment of the limit $\zeta_{-1} \rightarrow 0$. It is for this reason that UMAP’s learning-rate schedule must be annealed to zero (Figs. 1(c,d)). The repulsion shape (Eq. 12), by contrast, remains everywhere positive and satisfies Proposition 4.2: f_r^U tends to 0 as $\zeta \rightarrow \infty$ and to ∞ as $\zeta \rightarrow 0$ (Fig. 1(e)).

A comparison of the attraction shapes across the several algorithms reveals that only UMAP and t-SNE must contend with the predicament of $f_a < -1$ (and, consequently, $\zeta_{-1} > 0$ at $\lambda = 1$; Fig. 1(f)). t-SNE resolves this by weighting the updates with the corresponding p_{ij} values, whereas UMAP relies instead upon learning-rate annealing. Methods such as Neg-t-SNE and PaCMAP possess f_a that lies natively within $[-1, 0]$ (and thus satisfy Proposition 4.1 for $\lambda \in [0, 1]$ with $\zeta_{-1} = 0$); PaCMAP’s weighted f_a (which remains below 0.5 throughout) forestalls, furthermore, any flips during attraction. The repulsion shapes, for their part, indicate that UMAP alone must accommodate large values at small distances (Fig. 1(g)); the shape is unbounded, though its values are frequently clipped in the course of optimization). For further particulars concerning these and related methods, see Appendix G.

We exhibit the consequences of these attraction and repulsion choices for the MNIST embedding by imposing separate constant learning rates upon attraction (λ_a) and repulsion (λ_r) and varying their values (Figs. 1(h,i)). For UMAP, the lower value of λ_a (< 0.5 , and consequently $\zeta_{-1} \simeq 0$) is to be preferred in the interest of a superior embedding (trustworthiness (T) (Venna and Kaski, 2001), a measure of structure preservation, in Fig. 1(h)) and of clustering (the silhouette score (SIL) (Rousseeuw, 1987), a measure of cluster separation, in Fig. 1(i)), while the entire range of λ_r ($\in [0, 1]$) remains admissible. For NEG-t-SNE and PaCMAP, for which $\zeta_{-1} = 0$, the whole range of parameters proves efficacious. This corroborates the contention that UMAP depends upon driving ζ_{-1} close to 0 for a satisfactory embedding and clustering—an end it attains in practice through learning-rate annealing—whereas $\zeta_{-1} = 0$ obtains

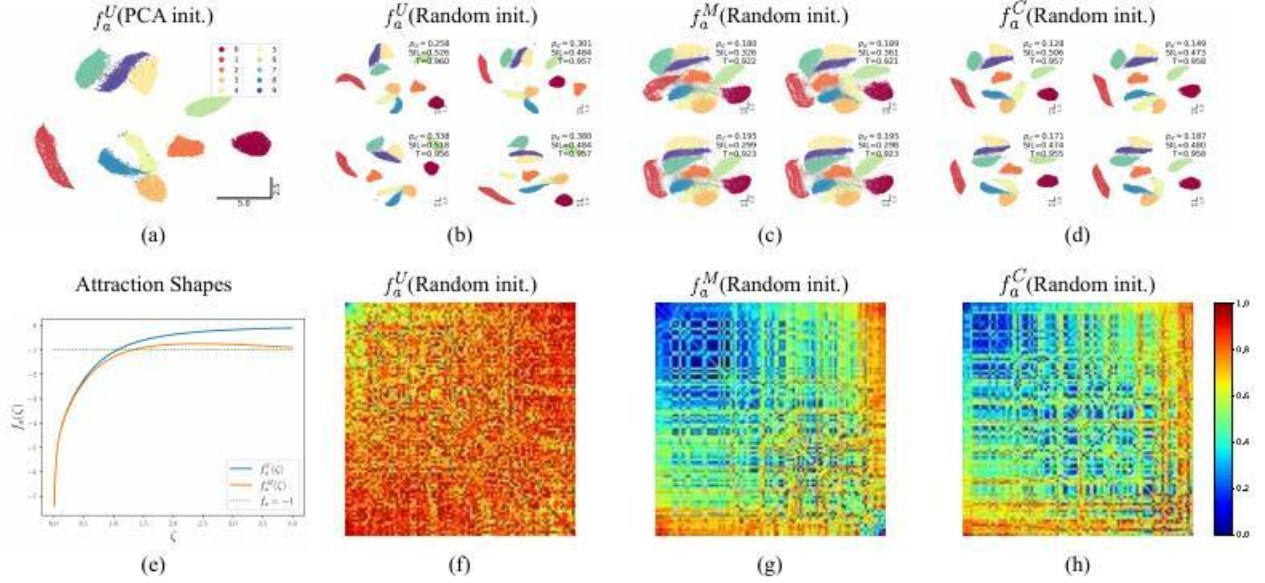


Figure 2: The effect of random initialization upon differing attraction shapes for the MNIST dataset. (a) Mapping obtained via PCA. (b-d) The four mappings exhibiting the lowest Procrustes distance (p_d) from the embedding in (a), for the (b) UMAP, (c) modified, and (d) composite attraction shapes. (e) The default UMAP and modified attraction shapes. (f-h) Procrustes matrices derived from 100 runs of the (f) UMAP (0.78 ± 0.13), (g) modified (0.49 ± 0.21), and (h) composite attraction (0.50 ± 0.20) shapes. The diagonal (i, i) entries of the Procrustes matrix are ordered by Procrustes distance (p_d) from (a), while the off-diagonal (i, j) entries correspond to the p_d obtaining between the i^{th} and j^{th} mappings. The matrices, together with the (mean $p_d \pm \text{std}$) values, attest that UMAP’s embeddings are not self-similar, whereas the modified and composite attraction shapes foster a scale-invariant structure.

by default in the newer algorithms. For further discussion, results, and illustrations of these methods across differing datasets, see Section 6 and Appendices B, C, and I.

5.1 Improving UMAP’s Consistency under Random Initialization

The consistency of UMAP’s embeddings is contingent upon a judicious initialization [Kobak and Linderman, 2021; Wang et al., 2021]. Ordinarily, principal component analysis (PCA) of the data, or a spectral decomposition of the high-dimensional graph, serves to initialize the embedding, thereby yielding consistent mappings notwithstanding the various sources of stochasticity at play. Under random initialization, by contrast, clusters frequently fail to form altogether, or else form in an arbitrary orientation upon each successive execution of the algorithm. Should the initial distance between a pair of points (nearest neighbors in the high dimension) be large, the attractive forces prove too feeble to draw them together. This phenomenon—known as near-sightedness [Wang et al., 2021]—is manifest in the attraction shape, wherein $|f_a^U|$ dwindles toward zero as the distance grows (that is, $\lim_{\zeta \rightarrow \infty} |f_a^U(\zeta)|\zeta = 0$).

One may induce a countervailing “far-sightedness” in the mapping by augmenting attraction at large distances, thereby enabling distant neighbors to approach one another. To put this hypothesis to the test, we modify the attraction shape of UMAP so as to intensify the attractive force:

$$f_a^M = f_a^U - \beta\zeta, \quad (13)$$

where β is a parameter regulating the strength of the appended term (we adopt $\beta = 0.2$; Fig. 2(e)). This addition to the attraction shape is tantamount to the introduction of a regularizer into the attractive term of the loss function (namely, $\mathcal{L}^M = \mathcal{L} + \sum_{i,j} \beta/3 \|y_i - y_j\|_2^2$). Beyond attracting pairs situated at remote distances, this device promotes an intermixing of points that abets convergence under random initialization (in a manner reminiscent of the early-exaggeration phase in t -SNE). In (13) we elected the simplest of linear corrections; other functions, such as $\log \zeta$ or ζ^p ($p \in \mathbb{R}_{\geq 0}$), might equally serve.

We also consider a composite attraction shape:

$$f_a^C = \begin{cases} f_a^M, & \text{epoch} \leq 100 \\ f_a^U, & \text{otherwise} \end{cases}. \quad (14)$$

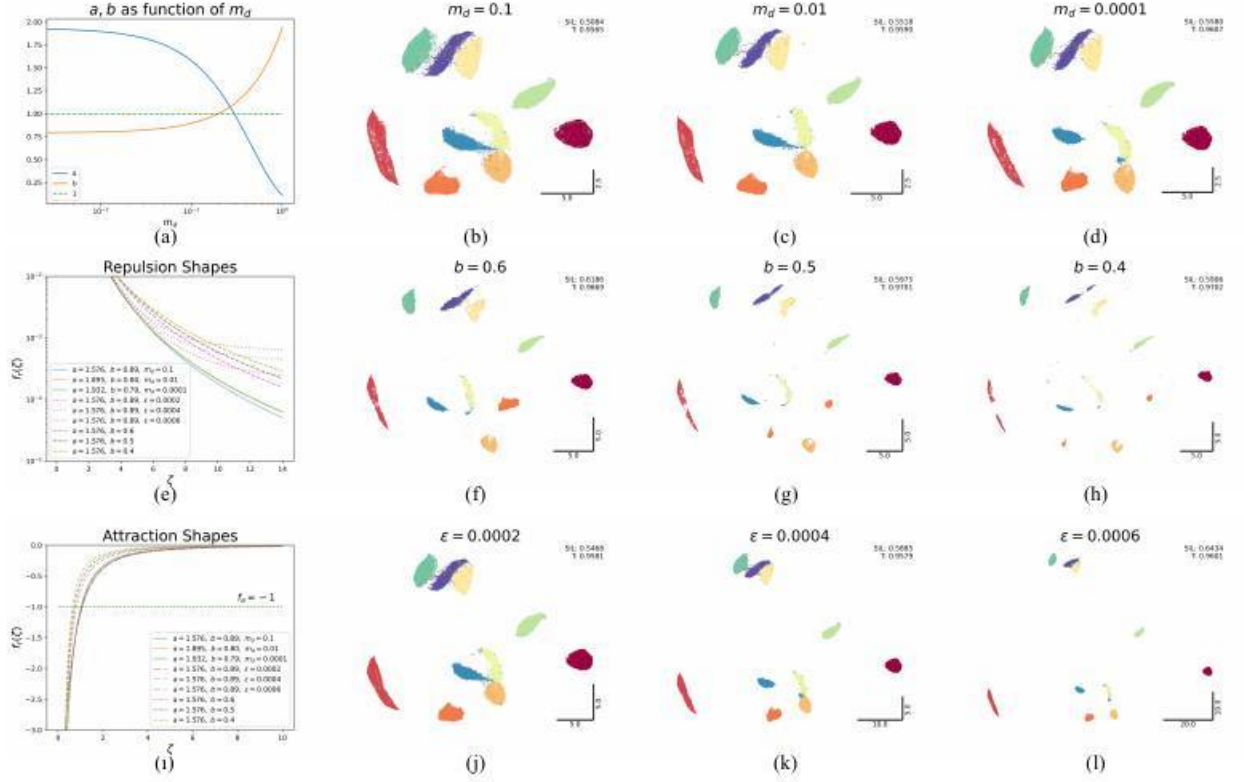


Figure 3: The control of inter-cluster distances on the MNIST dataset. (a) Computing a, b by varying the low-dimensional distance m_d curtails the scope for exploration. (b-d) UMAP output obtained by setting m_d to 0.1, 0.01, and 0.0001, respectively, evinces scant improvement in the compactness of the clusters. (e) Repulsion shapes for differing parameters. (f-h) Amplifying repulsion by explicitly varying b yields more compact clusters and precipitates new ones that were otherwise absent. (i) Attraction shapes across varying parameters. (j-l) Amplifying repulsion by adding a small positive value (ϵ) to the repulsion shape enlarges the inter-cluster distance.

The composite shape endeavors to expunge any distortions occasioned by f_a^M by reverting to the original UMAP. In what follows, we expound the effects that these modified and composite attraction shapes exert upon DR proceeding from random initialization.

We first generated a PCA-initialized embedding of the MNIST dataset (Figure 2(a)). We thereafter produced embeddings by way of random (Gaussian) initialization for each shape, iterating the experiment one hundred times. To render the results quantitative, we invoke Procrustes analysis (Gower 1975), which aligns two point clouds under scaling, translation, rotation, and reflection (for the details, see Appendix D.1). Here we align the randomly initialized embeddings to their PCA-initialized counterpart and characterize their separation by means of the Procrustes distance (p_d). Figure 2(b) presents the four embeddings of lowest p_d : while the cluster shapes remain consistent, their placements do not. The outputs arising from the modified and composite attraction shapes (Figs. 2(c) and (d), respectively) exhibit a markedly improved consistency in the placement of clusters.

To characterize the placements more finely, we consider the Procrustes matrix, whose diagonal is ordered by p_d from the PCA-initialized mapping and whose off-diagonal values register the p_d obtaining between two randomly initialized mappings (for the details, see Appendix D.1); this quantification is analogous to the similarity matrix (Foote 1999). The embeddings engendered by the default UMAP attraction shape bear little resemblance to one another (Fig. 2(f)), whereas the modified (Fig. 2(g)) and composite (Fig. 2(h)) shapes exhibit a pronounced mutual similarity. This attests that UMAP encodes a unique structure irrespective of the initialization; the realization of that structure in the low dimension may nonetheless prove deficient on account of the meagre attraction subsisting at longer distances.

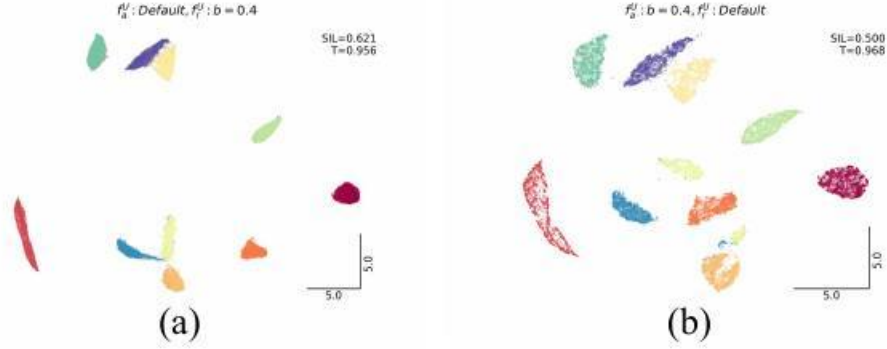


Figure 4: Embedding of the MNIST dataset obtained with (a) the default attraction shape but a repulsion shape at $b = 0.4$, and (b) the default repulsion shape but an attraction shape at $b = 0.4$. The former reproduces the very clusters of the default UMAP, albeit with heightened compactness; the latter develops novel structures within the clusters and precipitates clusters anew.

5.2 Cluster Formation and Compactness

The foremost controllable parameter bearing upon cluster formation in UMAP is the minimum-distance parameter m_d (by way of Eq. 2). Varying m_d , however, constrains the exploration of differing values of a and b (Fig. 3(a)); consequently, a reduction of m_d often issues in embeddings that confer no additional benefit (Figs. 3(b-d)). The decisive factor here is the circumscribed influence that a variation of m_d exerts upon the repulsion shape (Fig. 3(e)). As an alternative, we may vary the values of a and b explicitly. Diminishing a intensifies repulsion, yet it attenuates attraction at a still swifter rate—thereby aggravating the near-sightedness. Diminishing b , by contrast, affords finer control (Figs. 3(e,i)). Figures 3(f-h) chart the growing inter-cluster distance and the fragmentation of pre-existing clusters as b is varied to 0.6, 0.5, and 0.4, respectively. This fragmentation arises from the increasingly heavy-tailed character the kernel assumes as b decreases (heavy-tailed kernels beget smaller and more distinct clusters [Van der Maaten and Hinton, 2008; Kobak et al., 2019]; see Kobak et al. (2019) for an account of how varying b in UMAP modulates the tail). Although this stratagem partitions all ten MNIST labels into discrete clusters, the respective contributions of attraction and repulsion resist ready disentanglement: altering b amplifies repulsion more than altering m_d does, yet it likewise reshapes the attraction profile.

To quantify the effect of repulsion in isolation, we may hold attraction fixed while varying the former alone. To this end, we modify the repulsion shape through the addition of a small positive value (ε):

$$f_r^M = f_r^U + \varepsilon, \quad (15)$$

while holding the values of a and b constant—a manoeuvre that, in effect, introduces a regularizer into the repulsive term of the loss function (namely, $\mathcal{L}^M = \mathcal{L} - \sum_{i,j} \varepsilon/2 \|y_i - y_j\|_2^2$). By this means we may procure a stronger repulsion than was hitherto attainable (Fig. 3(e)). Figs. 3(j-l) attest that, as ε grows, the inter-cluster distances grow in concert. The clustering properties, however, closely resemble those obtained by varying m_d , and we secure a loose separation of all the labels as ε increases. On the whole, the parameter ε leaves the attraction shape unaffected, and varying m_d effectively retraces much the same attraction shapes (Fig. 3(i))—an observation intimating that cluster formation is governed predominantly by attraction.

To dispel the mystery surrounding cluster formation, we alter either the attraction shape or the repulsion shape individually—leaving the other at its default—by the simple expedient of setting b to 0.4 (Fig. 4). The default attraction is unable to elicit new structures or clusters within the embedding, whereas the heightened repulsion ($b = 0.4$) yields clusters smaller than those of the original UMAP (Fig. 4(a)). When, conversely, attraction is heightened by setting $b = 0.4$ against the default repulsion, the embedding discloses additional structures within each cluster (Fig. 4(b)), and certain of the erstwhile clusters even fission into smaller ones. This demonstrates that attraction is the agent of cluster formation, while repulsion serves to render the clusters more compact (accomplishing this, according to the repulsion shape, either by begetting smaller clusters or by enlarging the inter-cluster distance). We show in Appendix G.3 that LocalMAP embeddings likewise exploit this interplay of attraction and repulsion in order to separate the clusters.

6 Comparison to NEG- t -SNE

The setting of a and b to 1 is a commonplace across a variety of dimensionality reduction algorithms; this particular choice renders the gradients of many such algorithms stable (or bounded), and, in turn, eases the task of optimization. Recently, Damrich et al. (2023) explored this setting for UMAP and proposed Neg- t -SNE as a remedy—one that, besides possessing a stable gradient, furnishes a more compact clustering even under a constant learning rate of 1. Parametric UMAP (Sainburg et al. 2021), for its part, at first employed the UMAP loss formulation, but subsequently adopted a numerically stable, modified cross-entropy loss function (Shi et al. 2023) equipped with a logsigmoid kernel. An analysis of both reveals that the two approaches converge upon the selfsame formulation from disparate points of departure, which prompts the following observation:

Proposition 6.1. *Neg- t -SNE is Parametric UMAP with $a = 1$ and $b = 1$.*

With $a = 1$ and $b = 1$, the attraction and repulsion shapes of UMAP are given by $f_a^U = -2/(1 + \zeta^2)$ and $f_r^U = 2/(\zeta^2(1 + \zeta^2))$, respectively. The attraction shape thereby becomes bounded within $[-2, 0]$, with $f_a^U(0) = -2$ (Fig. 1(f)), whereas the repulsion shape remains essentially unaltered (that is, unbounded as $\zeta \rightarrow 0$; Fig. 1(g)). Since $f_a^U < -1$ as $\zeta \rightarrow 0$, it follows—by Proposition 4.1 and the discussion advanced in Section 5—that this unity case continues to demand learning-rate annealing.

Damrich et al. (2023) examined UMAP’s negative-sampling loss function through the lens of contrastive embedding (CE) and inferred that the effective kernel of UMAP is $1/\zeta^2$. Within the CE framework, the authors introduced NEG- t -SNE by supplanting the kernel with $1/(1 + \zeta^2)$. Transposed back into the UMAP formalism, this yields the low-dimensional affinity function $q_{ij}^N = 1/(2 + \zeta^2)$, whence the attraction and repulsion shapes follow as $f_a^N = -2/(2 + \zeta^2)$, and $f_r^N = 2/((1 + \zeta^2)(2 + \zeta^2))$, respectively (Figs. 1(f,g)). The attraction shape is bounded within $[-1, 0]$ and satisfies Proposition 4.1: any $\lambda \in [0, 1]$ engenders contraction and averts the oscillation between expansion and contraction. NEG- t -SNE is, in consequence, less sensitive to learning-rate annealing, and its clusters present a less fuzzy aspect even under a constant $\lambda = 1.0$. The repulsion shape of NEG- t -SNE, moreover, is likewise bounded—within $[0, 1]$ —and does not diverge to infinity as $\zeta \rightarrow 0$, so that fewer points depart the clusters when sampled at random. Whereas Damrich et al. (2023) appealed solely to the bounded repulsive forces of NEG- t -SNE and the unbounded ones of UMAP to account for this disparity, our analysis in Fig. 1 makes clear that the attraction shape is equally answerable for the stability and compactness of the clusters. For further discussion of NEG- t -SNE, accompanied by illustration, see Appendix F. Taken in conjunction with the discussion attending Fig. 4 this intimates that attraction and repulsion shapes drawn from different methods may be commingled—pairing, for instance, the attraction shape of UMAP with the repulsion shape of NEG- t -SNE. We furnish a handful of such examples through experiments on UMAP, NEG- t -SNE, and PaCMAP in Appendix J.

7 Discussion and Conclusion

In the present work, we investigated the relationship subsisting between the attractive and repulsive forces and cluster formation. Although it is well established that attractive forces draw similar points together while repulsive forces drive dissimilar ones apart, the precise mechanisms by which these forces operate had hitherto eluded thorough study. Here we have demystified much of the dynamics underpinning cluster formation. While our attention in the main text was trained upon UMAP and NEG- t -SNE, the results generalize to dimensionality reduction at large (several further algorithms are treated in Appendix G).

We treated the attraction and repulsion coefficients as functional mappings and analyzed them in terms of their respective shapes. The characterization of these shapes (Propositions 4.1 and 4.2) laid bare a counterintuitive result: the attractive forces of UMAP occasion repulsion—an expansion of distance—at shorter distances (Fig. 1(b)); that is, rather than drawing neighboring points together, they thrust them apart. We conclude that UMAP’s learning-rate schedule serves to mitigate this phenomenon.

We further scrutinized the initialization of the low-dimensional embedding. Similar points that commence farther apart experience feeble attraction and never contract adequately. Our formalism supplies a means of influencing attraction at larger distances through the addition of supplementary terms to the attraction shape. This modification issued in outputs of greater consistency under random initialization—a finding that lends us confidence that UMAP encodes a unique structure, the numerical expedients serving merely to attain it more expeditiously.

An analysis of the repulsion shape disclosed that stronger repulsion begets a larger inter-cluster distance. To this end, we appended a small positive value to the shape, thereby achieving a heightened repulsion unattainable by the traditional device of varying the minimum-distance parameter m_d .

github.com, merged on April 26, 2022

Finally, we explored the relation between NEG- t -SNE and UMAP. In the particular case of $a = 1$ and $b = 1$, the salient distinction is that NEG- t -SNE, in its default form, furnishes bounded attraction ($\in [-1, 0]$) and repulsion ($\in [0, 1]$) shapes; both propositions are thus satisfied irrespective of the learning rate, whence the method displays diminished sensitivity to a fixed learning rate ($\in [0, 1]$). Varying a and b in NEG- t -SNE, however, introduces the very numerical challenges besetting UMAP. We demonstrated, in addition, the equivalence of NEG- t -SNE and Parametric UMAP.

These insights into attraction–repulsion dynamics proffer novel instruments for the optimization of dimensionality reduction algorithms. Beyond this, the intimate connection between dimensionality reduction and contrastive learning (Damrich and Hamprecht 2021, Hu et al. 2023) suggests that our approach may equally serve to advance representation learning. Considered in the aggregate, our work aspires to render embeddings and their interpretations more principled, more consistent, and more reliable, and thereby to guide future inquiry.

Software and Data

All of the data employed in this paper are publicly available. The MNIST dataset may be obtained at yann.lecun.com. Fashion-MNIST is available at github.com. Single-cell transcriptomes data is available at github.com.

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Appendix

In Appendix [A](#) we set out the requisite derivations pertaining to the attraction and repulsion shapes of UMAP. Extending our discussion from the main text, we revisit the use of a constant learning rate in Appendix [B](#). Thereafter, in Appendix [C](#) we examine the implications of improved attraction and repulsion for large language model embeddings. We formally define the Procrustes distance and matrix in Appendix [D.1](#) and explore additional datasets for the random-initialization experiment in Appendix [E](#). We prolong the discussion of NEG- t -SNE and UMAP in Appendix [F](#). In Appendix [G](#) we canvass alternate dimensionality reduction methods, and, for the sake of completeness, we treat the construction of the high-dimensional graph in these methods in Appendix [H](#). In Appendix [I](#) we present the detailed results of varying λ_a and λ_r (from Fig. [1](#)) for UMAP, NEG- t -SNE, and PaCMAP across differing datasets. We exhibit the effect of mixing and matching the shapes drawn from different algorithms in Appendix [J](#). Finally, we supply implementation details in Appendix [K](#).

A Proofs and Derivations

A.1 Proof of Proposition [4.1](#)

Proof. We subtract Eq. [\(7\)](#) from Eq. [\(8\)](#) and take a norm:

$$\|y_i^{t+1} - y_j^{t+1}\| = |1 + 2\lambda f_a| \|y_i^t - y_j^t\|. \quad (16)$$

This distance contracts as long as $|1 + 2\lambda f_a| < 1$, i.e., provided

$$-1 < \lambda f_a < 0, \quad (17)$$

□

A.2 Proof of Proposition [4.2](#)

Proof. We subtract Eq. [\(9\)](#) from Eq. [\(10\)](#) and take a norm:

$$\|y_i^{t+1} - y_j^{t+1}\| = |1 + \lambda f_r| \|y_i^t - y_j^t\|. \quad (18)$$

This distance increases when $|1 + \lambda f_r| > 1$, i.e.,

$$\lambda f_r < -2 \text{ or } f_r > 0. \quad (19)$$

From the definition of f_r , the latter suffices. □

A.3 Attraction Shape:

To derive the attraction shape we avail ourselves of a general form of the low-dimensional affinity function, namely $q_{ij} = (\gamma + a\|y_i - y_j\|_2^{2b})^{-1}$, which reduces to UMAP for $\gamma = 1$ and to NEG- t -SNE for $\gamma = 2$. The attractive force is then given by

$$\begin{aligned} \nabla_{y_i} \log q_{ij} &= -\nabla_{y_i} \log (\gamma + a(\|y_i - y_j\|_2^2)^b) \\ &= -\frac{1}{\gamma + a(\|y_i - y_j\|_2^2)^b} \nabla_{y_i} (\gamma + a(\|y_i - y_j\|_2^2)^b) \\ &= -\frac{1}{\gamma + a(\|y_i - y_j\|_2^2)^b} ab(\|y_i - y_j\|_2^2)^{b-1} \nabla_{y_i} \|y_i - y_j\|_2^2 \\ &= -\frac{2ab(\|y_i - y_j\|_2^2)^{b-1}}{\gamma + a(\|y_i - y_j\|_2^2)^b} (y_i - y_j). \end{aligned} \quad (20)$$

Upon defining $\zeta = \|y_i - y_j\|_2$, the first term yields the attraction shape as:

$$f_a(\zeta) = -\frac{2ab\zeta^{2(b-1)}}{\gamma + a\zeta^{2b}} \quad (21)$$

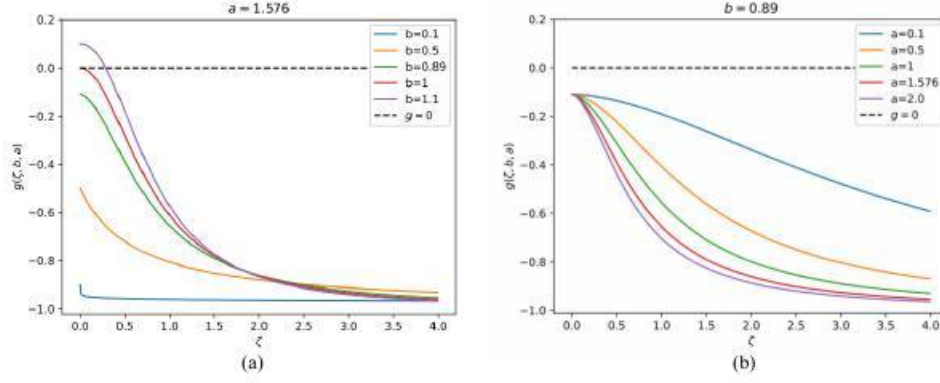


Figure 5: The values assumed by $g(\zeta, b, a)$ for (a) a held fixed at 1.576 and (b) b held fixed at 0.89.

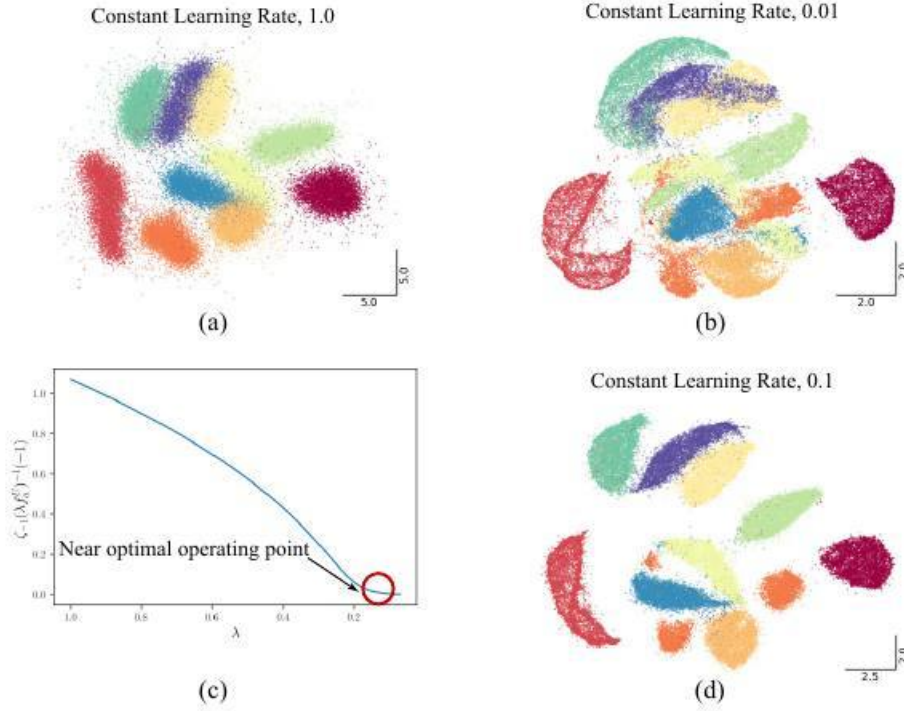


Figure 6: The effect of a constant learning rate upon embeddings. (a) When the learning rate is excessive ($\lambda = 1.0$), the embeddings turn diffuse (on account of the elevated value of ζ_{-1}). (b) When the learning rate is too low ($\lambda = 0.01$), no clusters form (the strengths of attraction and repulsion being too feeble). (c) ζ_{-1} diminishes nonlinearly as the learning rate decreases; the objective of the algorithm is to reduce ζ_{-1} while preserving effective levels of attraction and repulsion. (d) Distinct and compact clusters emerge at a constant, near-optimal learning rate of $\lambda = 0.1$.

A.4 Condition for strictly increasing f_a^U :

Its behavior as a function of distance may be characterized by computing the derivative of f_a^U :

$$\frac{df_a^U(\zeta)}{d\zeta} = -\frac{2ab\zeta^{2b-3}}{\gamma + a\zeta^{2b}} \left((b-1) - \frac{ab\zeta^{2b}}{\gamma + a\zeta^{2b}} \right). \quad (22)$$

This leads to a strictly increasing condition ($\frac{df_a^U}{d\zeta} > 0$),

$$g(\zeta, b, a) < 0, \quad (23)$$

where $g(\zeta, b, a) = b - 1 - \frac{ab\zeta^{2b}}{\gamma + a\zeta^{2b}}$. This inequality remains valid so long as $0 < b \leq 1$ (as may be established from the derivative and asymptotes of g). Figure 5 displays the values of g for differing b and a . As b rises above 1, the inequality (23) ceases to hold.

A.5 Repulsion Shape:

The repulsive force, expressed by means of the general form of the low-dimensional affinity, is given by

$$\begin{aligned}\nabla_{y_i} \log(1 - q_{ij}) &= \nabla_{y_i} \log \left[1 - \frac{1}{\gamma + a(\|y_i - y_j\|_2^2)^b} \right] \\ &= \frac{\gamma + a(\|y_i - y_j\|_2^2)^b}{(\gamma - 1) + a(\|y_i - y_j\|_2^2)^b} \nabla_{y_i} \left[1 - \frac{1}{\gamma + a(\|y_i - y_j\|_2^2)^b} \right] \\ &= \frac{1}{\gamma - 1 + a(\|y_i - y_j\|_2^2)^b} \frac{ab(\|y_i - y_j\|_2^2)^{b-1}}{\gamma + a(\|y_i - y_j\|_2^2)^b} \nabla_{y_i} \|y_i - y_j\|_2^2 \\ &= \frac{2ab(\|y_i - y_j\|_2^2)^{b-1}}{(\gamma - 1 + a(\|y_i - y_j\|_2^2)^b)(\gamma + a(\|y_i - y_j\|_2^2)^b)} (y_i - y_j).\end{aligned}\quad (24)$$

The first term yields the repulsion shape as:

$$f_r(\zeta = \|y_i - y_j\|_2) = \frac{2ab\zeta^{2(b-1)}}{(\gamma - 1 + a\zeta^{2b})(\gamma + a\zeta^{2b})}.\quad (25)$$

Generally, $f_r > 0$.

A.6 Loss functions due to modified attraction and repulsion shape:

The cost function of the attractive term, incorporating the modification of Eq. (13), is given by

$$-\int (f_a^U(\|y_i - y_j\|_2) - \beta\|y_i - y_j\|_2)(y_i - y_j)dy_i = -\log(q_{ij}) + \frac{\beta}{3}\|y_i - y_j\|_2^3,\quad (26)$$

whereas the repulsive term attributable to Eq. (15) is given by

$$-\int (f_r^U(\|y_i - y_j\|_2) + \epsilon)(y_i - y_j)dy_i = -\log(1 - q_{ij}) - \frac{\epsilon}{2}\|y_i - y_j\|_2^2.\quad (27)$$

In both instances the supplementary term functions as a regularizer, expressed simply in terms of norms. When, however, we append this term directly to the attraction and repulsion shapes, the office performed by each term becomes readily explicable: for the attractive term it augments far-sightedness, while for the repulsive term it contributes a constant repulsive coefficient.

B More on UMAP's Learning Rate

As we discussed in Sections 5 and 6 of the main text, UMAP is held to require learning-rate annealing (Fig 6). By way of explanation, in Section 5 we introduced the notion of the minimum distance for contraction (ζ_{-1}) and established that a reduction of this value through learning-rate annealing gives rise to compact clusters. Subsequently, in Section 6 we juxtaposed the attraction shapes of UMAP (for $a = 1.0$ and $b = 1.0$) and Neg- t -SNE and explained that Neg- t -SNE withstands a constant learning rate of 1.0 more successfully than UMAP, inasmuch as its attraction shape resides within $[-1, 0]$ whereas that of UMAP lies within $[-2, 0]$. Pursuing the same logic, we demonstrated that UMAP can withstand a constant learning rate of 0.5 by constraining its attraction shape to remain within $[-1, 0]$ (Figs. 11(f,h)).

The embeddings are nonetheless superior when the learning rate is annealed (across a broad range of the parameters a and b). This is because the ultimate aim of the algorithm is to reduce ζ_{-1} to zero, an outcome that transpires only as the learning rate itself approaches zero; absent such annealing, the embedding grows diffuse (Fig 6(a)). Conversely, should the learning rate be too low from the outset, the strength of both attraction and repulsion proves too feeble, and no clear clusters can form (Fig 6(b)). Analysis of the ζ_{-1} -versus- λ curve (Fig 6(c)) reveals a near-optimal point at $\lambda = 0.1$, at which ζ_{-1} is low while attraction and repulsion retain considerable strength. Fixing the constant learning rate at 0.1, we accordingly obtain compact clusters delineated by clear boundaries (Fig 6(d)).

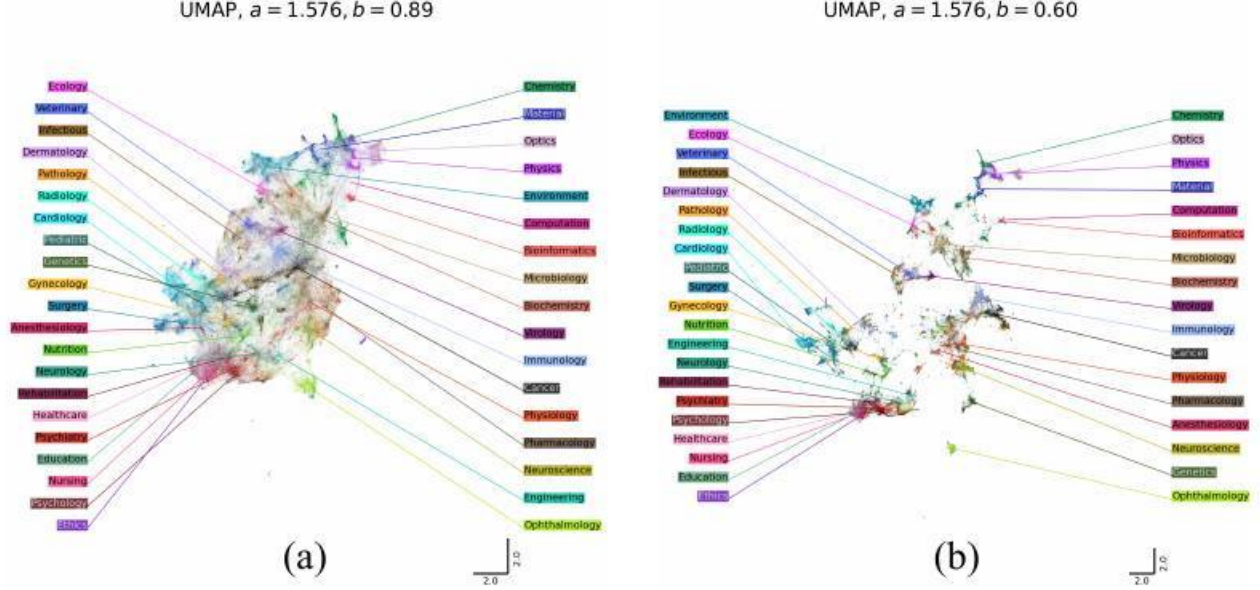


Figure 7: Visualization of 300K abstracts drawn from the PubMed dataset. (a) The default UMAP ($a = 1.576, b = 0.89$) yields a mapping in which the labels overlap one another, whereas (b) UMAP endowed with improved repulsion and attraction ($a = 1.576, b = 0.60$) affords superior cluster formation and non-overlapping labels. Both mappings are labeled to facilitate comprehension.

C Large Language Model Embedding

Heightened attraction and repulsion frequently give rise to superior embeddings. To demonstrate this, we selected at random 300,000 feature samples—extracted by means of PubMedBERT (Gu et al. [2021])—bearing known labels from the PubMed dataset (González-Márquez et al. [2024]). The default UMAP furnishes a crowded structure in which the labels (colors) overlap (Fig. 7a); in the absence of labels, one might well be perplexed by the embeddings, since explicit clusters are wanting. By adjusting b to 0.6 (in the manner discussed in Section 5.2), however, we intensify repulsion while simultaneously abetting cluster formation through improved attraction. The resulting mapping is amenable to exploration even without explicit labeling (Fig. 7b)). The improvement of the embedding may be quantified through the k -NN accuracy ($k = 10$): on a hold-out set of 10,000 points, the k -NN accuracy rose from 49.2% to 55.48% upon reducing b to 0.6.

D Metrics

D.1 Procrustes Distance and Procrustes Matrix

The Procrustes distance (Gower [1975]) gauges the similarity between two point clouds $\{x\}$ and $\{y\}$ under linear transformations—namely translation, scaling, and rotation. Operationally, we hold the former set fixed and vary the latter until the two attain maximal alignment. Letting $\{y'\}$ denote the transformation of $\{y\}$ that realizes this objective, the Procrustes distance is then given by

$$pd(\{x\}, \{y\}) = \sqrt{\sum_k (x_k - y'_k)^2}. \quad (28)$$

The Procrustes distance is a linear measure that has proven serviceable across a variety of settings (McInnes et al. [2018]; Islam and Fleischer [2022]; Kotlov et al. [2024]).

Here we enlist the Procrustes distance to gauge the consistency of the embedding under random initialization. Let $\{x\}_p$ denote a reference embedding (obtained via PCA initialization in our experiments), and let $X_r = \{\{x\}_i | i = 1, 2, 3, \dots, N\}$ denote a set of N embeddings procured from random initialization. The similarity of the embeddings may be quantified by computing the mean and standard deviation of the strictly lower-triangular values of the matrix P

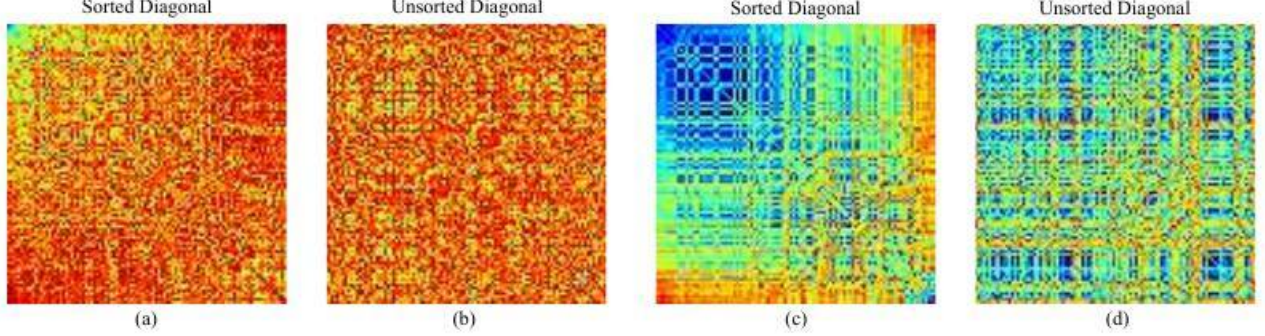


Figure 8: The effect exerted upon the visualization of the Procrustes matrix by sorting its diagonal. (a) The Procrustes matrix reproduced from [2]f) for the default UMAP attraction shape, the diagonal being ordered by the p_d of a sample embedding under PCA initialization. (b) The same data as in (a), save that the diagonal is left unsorted (or randomly sorted). (c) The Procrustes matrix reproduced from [2]g) for the modified attraction shape, the diagonal again being ordered by the p_d of a sample embedding under PCA initialization. (d) The same data as in (c), save that the diagonal is left unsorted.

(reported as mean $\pm$ std in Figs. [2] [9] and [10], with

$$\text{mean} = \frac{2}{N(N-1)} \sum_{i,j(i>j)} P_{i,j}, \text{ and } \text{std} = \sqrt{\sum_{i,j(i>j)} \frac{2(P_{i,j} - \text{mean})^2}{N(N-1)}} \quad (29)$$

The indices of X_τ may be ordered such that $p_d(\{x\}_i, \{x\}_p) \leq p_d(\{x\}_{i+1}, \{x\}_p)$, whereupon the diagonal values of the Procrustes matrix are given by

$$P_{i,i} = p_d(\{x\}_i, \{x\}_p) \quad (30)$$

and the off-diagonal values are given by

$$P_{i,j} = p_d(\{x\}_i, \{x\}_j). \quad (31)$$

Numerically, the sorting of the diagonal contributes little; visually, however, the ordering brings to light the underlying self-similarity of the embeddings. In Fig. [8]a), for instance, the sorted diagonal reveals that similar embeddings congregate in the upper-left region of the matrix. When a modified attraction shape is employed, the number of mutually similar points increases (as evinced by the enlarged blue region in Fig. [8]c)), betokening the presence of a metastable point within the embedding algorithm. When the diagonal is left unsorted, by contrast, this region vanishes and all sense of similar embeddings is forfeited (Fig. [8]b,d)).

D.2 Trustworthiness

The trustworthiness metric (Venna and Kaski, 2001) quantifies the fidelity with which local neighborhoods are preserved in the wake of dimensionality reduction:

$$T = 1 - \frac{2}{nk(2n - 3k - 1)} \sum_{i=1}^n \sum_{y_j \in \text{KNN}(y_i, k)} \max(0, r(i, j) - k) \quad (32)$$

where $\text{KNN}(y_i, k)$ denotes the k -NN graph in the embedding space and $r(i, j)$ the rank of x_j within the high-dimensional k -NN graph. In practice k is commonly set to 5, thereby assessing the preservation of each point's five nearest neighbors. In the interest of computational efficiency, we sampled 10,000 indices at random when reporting trustworthiness, and, when comparing distinct embeddings, we retained the same indices throughout.

D.3 Silhouette Score

Although the silhouette score (Rousseeuw, 1987) is properly directed at the assessment of clustering algorithms, we press it into service to evaluate label separation within the embeddings—that is, the degree to which the ground-truth labels have been clustered. The premise is that the embedding algorithms naturally engender clusters and ought to

separate the labels as far as possible. For a point y_i within a point cloud $\{y\}$, let a_i denote the mean distance from y_i to the other points sharing its label, and let b_i denote the minimal mean distance from y_i to points bearing any other label. The pointwise silhouette is accordingly given by

$$s_i = \frac{b_i - a_i}{\max\{a_i, b_i\}} \quad (33)$$

This value resides within $[-1, 1]$: a value close to 1 signifies that y_i sits comfortably within its own label cluster, one near 0 betokens a boundary point, and one close to -1 signals a failure of label separation. The overall silhouette score is

$$\text{SIL} = \frac{1}{N} \sum s_i. \quad (34)$$

We computed the silhouette score over the whole embedding (dispensing with random sampling) by means of Euclidean distances.

E Effect of Random Initialization in Additional Datasets

In the main text (Section 5.1) we presented results for the MNIST dataset alone. Here we replicate the same experiment on the Fashion-MNIST (FMNIST) (Xiao et al. 2017) and single-cell transcriptomes (Macosko et al. 2015) data (Figs. 9 and 10, respectively). The principal conclusion stands unaltered: modified and composite attraction shapes—those, that is, which augment attraction at large distances—markedly improve the consistency of reconstruction.

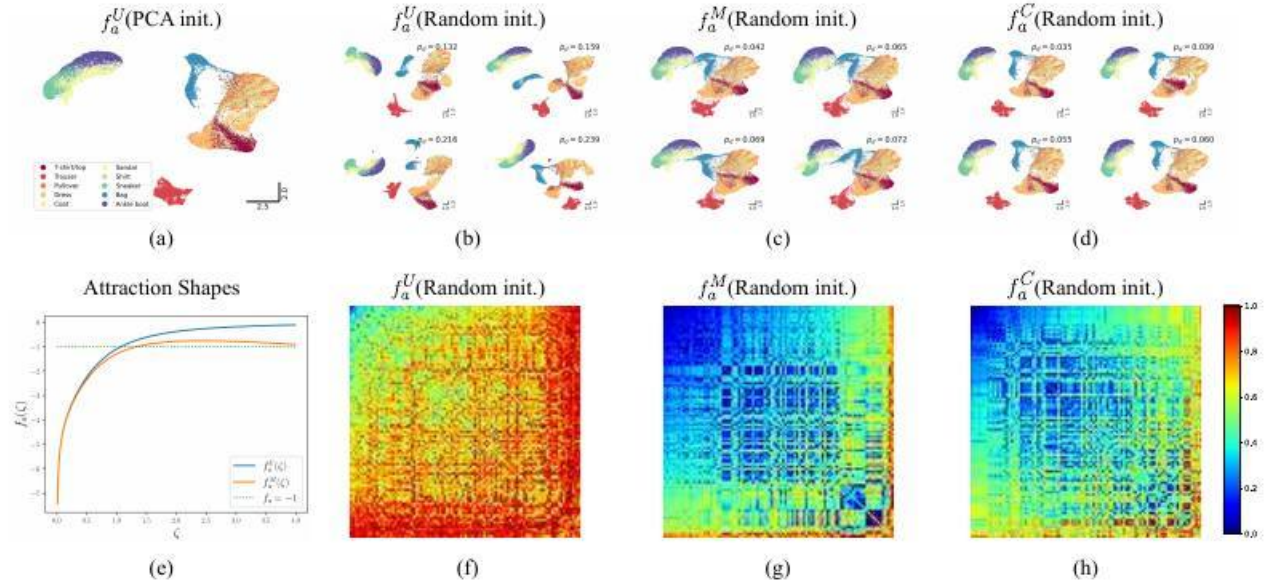


Figure 9: The effect of random UMAP initialization upon differing attraction shapes for the FMNIST data. (a) Mapping obtained via PCA, taken as the standard. (b-d) The four mappings exhibiting the lowest Procrustes distance (p_d) from the embedding in (a), for the (b) default, (c) modified, and (d) composite attraction shapes. (e) The default UMAP and modified attraction shapes. (f-h) Procrustes matrices derived from 100 runs of the (f) default (0.72 ± 0.15), (g) modified (0.38 ± 0.19), and (h) composite (0.43 ± 0.18) attraction shapes.

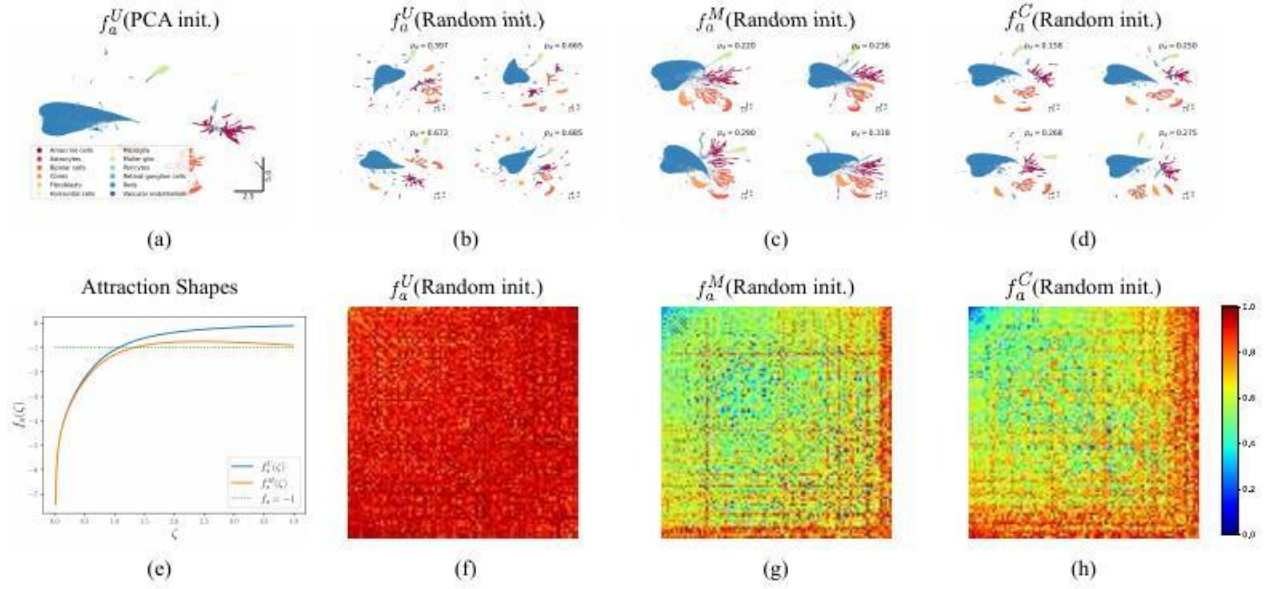


Figure 10: The effect of random UMAP initialization upon differing attraction shapes for the single-cell transcriptomes data. (a) Mapping obtained via PCA, taken as the standard. (b-d) The four mappings exhibiting the lowest Procrustes distance (p_d) from the embedding in (a), for the (b) default, (c) modified, and (d) composite attraction shapes. (e) The default UMAP and modified attraction shapes. (f-h) Procrustes matrices derived from 100 runs of the (f) default (0.91 ± 0.06), (g) modified (0.61 ± 0.13), and (h) composite (0.64 ± 0.15) attraction shapes.

F More on NEG- t -SNE

F.1 Comparison to UMAP

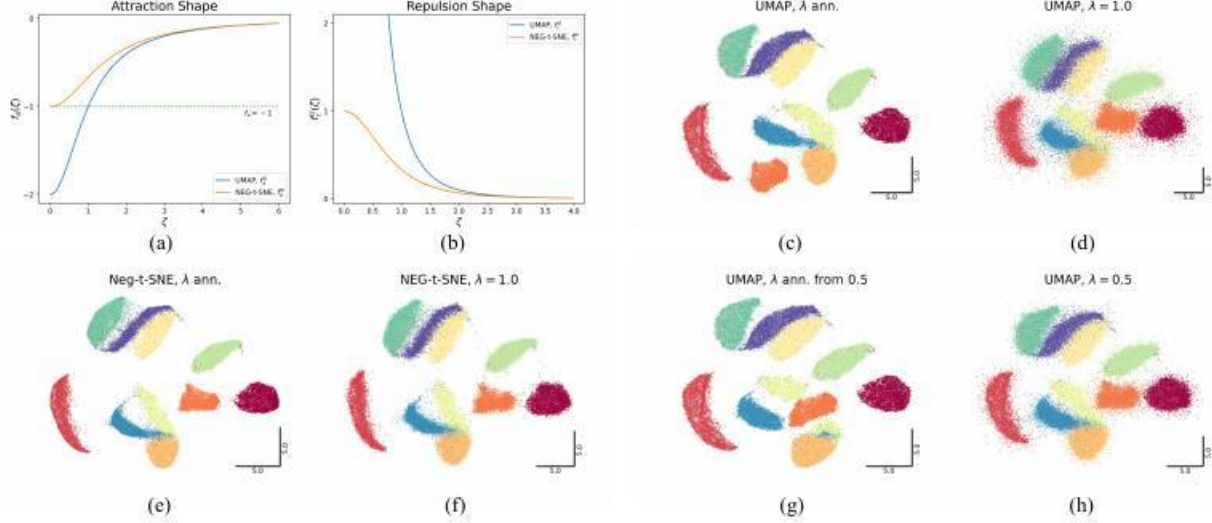


Figure 11: The sensitivity of UMAP and NEG- t -SNE to the learning rate on the MNIST dataset. (a) Attraction and (b) repulsion shapes for UMAP ($a = 1$, $b = 1$) and NEG- t -SNE. (c,d) UMAP proves highly sensitive to the learning rate λ , since $f_a^U < -1$ as the separation distance ζ diminishes; absent annealing, the clusters accordingly turn fuzzy. (e,f) NEG- t -SNE is less sensitive to λ , inasmuch as $f_a^N \in [-1, 0]$ throughout, so that the clusters remain less fuzzy even when left unannealed. (g,h) Confining f_a^U to $[-1, 0]$ by setting $\lambda = 0.5$ likewise attenuates the sensitivity to λ .

Figure 11 exhibits the shapes of UMAP and NEG- t -SNE alongside various MNIST embeddings. When the learning rate is annealed, UMAP and NEG- t -SNE alike yield comparable output (Figs. 11(c,e)). Under a constant learning rate of 1, however, UMAP presents a fuzzy structure whereas NEG- t -SNE presents one endowed with markedly cleaner boundaries (Fig. 11(d,f)). The discussion of Section 6 intimates that constraining f_a^U to lie within $[-1, 0]$ may plausibly yield less fuzzy clusters at fixed λ —a phenomenon we had already witnessed in Fig. 11(h,i), where UMAP afforded a superior embedding and clustering when the learning rate for attraction (λ_a) was $\lesssim 0.5$. A straightforward means of securing this is to initialize λ to 0.5, which satisfies Proposition 4.1 for all ζ . The resulting embeddings, displayed in Figs. 11(g) and (h), confirm that the clusters resemble those of NEG- t -SNE and that, under a constant $\lambda = 0.5$, they are—as foretold—less fuzzy than before, with sharper boundaries (Fig. 11(d,h)). A residue of points nonetheless persists beyond the clusters, owing to the characteristics of UMAP’s repulsion shape, which NEG- t -SNE remedies.

We may next introduce the parameters a and b into NEG- t -SNE (which amounts, in essence, to the formulation of Parametric UMAP; see Proposition 6.1). The affinity function thereupon becomes $q_{ij}^N = 1/(2 + a\zeta^{2b})$, and the attraction and repulsion shapes become $f_a^N = -\frac{2ab\zeta^{2(b-1)}}{2 + a\zeta^{2b}}$ and $f_r^N = \frac{2ab\zeta^{2(b-1)}}{(1 + a\zeta^{2b})(2 + a\zeta^{2b})}$, respectively. For $0 < b < 1$, both shapes grow unbounded as $\zeta \rightarrow 0$; NEG- t -SNE will thus confront numerical challenges akin to those of UMAP should a and b be varied, and the corresponding limitations carry over. One notable distinction is that, relative to UMAP, the attraction shape attains a lower minimum distance (ζ_{-1}) for the attraction. Although this may enhance cluster formation, it converges to zero more rapidly (an intensified near-sightedness as distance grows), potentially diminishing its efficacy for attraction over longer distances.

F.2 Comparison to Parametric UMAP

Parametric UMAP was originally trained under the original UMAP objective (Sainburg et al. 2021), though later work adopted a numerically stable, log-sigmoid-based modified cross-entropy loss (Shi et al. 2023). This modification renders Parametric UMAP and Neg- t -SNE (Damrich et al. 2023) equivalent (Proposition 6.1), an equivalence we establish below.

Proof of Proposition 6.1

Proof. Under this modification the kernel function becomes

$$q_{ij}^P = -\log(1 + a\|y_i - y_j\|_2^{2b}). \quad (35)$$

The attractive term is

$$-\text{logsigmoid}(q_{ij}^P) = -\log\left(\frac{1}{1 + \exp(-q_{ij}^P)}\right) \quad (36)$$

$$= -\log\left(\frac{1}{2 + a\|y_i - y_j\|_2^{2b}}\right) \quad (37)$$

$$= -\log(q_{ij}^N). \quad (38)$$

And the repulsive term is

$$\text{logsidmoid}(q_{ij}^P) - q_{ij}^P = \log\left(\frac{1}{2 + a\|y_i - y_j\|_2^{2b}}\right) + \log(1 + a\|y_i - y_j\|_2^{2b}) \quad (39)$$

$$= \log\left(\frac{1 + a\|y_i - y_j\|_2^{2b}}{2 + a\|y_i - y_j\|_2^{2b}}\right) \quad (40)$$

$$= \log\left(1 - \frac{1}{2 + a\|y_i - y_j\|_2^{2b}}\right) \quad (41)$$

$$= \log(1 - q_{ij}^N). \quad (42)$$

Both of these coincide with NEG-t-SNE endowed with explicit parameters a and b (whereas in Neg-t-SNE these are fixed at 1). $\square$

G Alternate Dimensionality Reduction Algorithms

The alternative algorithms we take up here employ the same kernel function as UMAP (with $a = 1$ and $b = 1$ in their low-dimensional weight):

$$q_{ij} = \frac{1}{1 + \|y_i - y_j\|_2^2}. \quad (43)$$

In this section we first expound the TriMap (Amid and Warmuth, 2019) algorithm. Although this algorithm rests upon triplets rather than pairwise interactions, it serves as a primer for the analysis of attraction and repulsion in a setting somewhat more involved than that of UMAP. This treatment is succeeded by Pairwise Controlled Manifold Approximation (PaCMAP) (Wang et al., 2021) together with its extension, Pairwise Controlled Manifold Approximation with Local Adjusted Graph (LocalMAP) (Wang et al., 2025), both of which recast TriMAP’s loss function so as to operate on pairwise interactions. We then turn to t -SNE (Van der Maaten and Hinton, 2008), append a brief note on SNE (Hinton and Roweis, 2002)—which employs an alternate kernel function—and finally close the section with a short discussion of multidimensional scaling (Borg and Groenen, 2007).

G.1 TriMap

TriMap (Amid and Warmuth, 2019) optimizes low-dimensional using a triplet loss

$$\mathcal{L}^T = \sum_{(i,j,k)} w_{ijk} \frac{1}{1 + \frac{q_{ij}}{q_{ik}}}, \quad (44)$$

where w_{ijk} denotes the weight of the triplet (y_i, y_j, y_k) , y_j belongs to the k -nearest neighbor set of y_i in the high dimension, and y_k is a distant point. Upon minimization, we anticipate that y_i and y_j attract one another while y_i and y_k repel one another. The update equations are

$$y_i^{t+1} = y_i^t + \lambda f_a^T(\zeta_1^t, \zeta_2^t)(y_i^t - y_j^t) + \lambda f_r^T(\zeta_1, \zeta_2)(y_i^t - y_k^t), \quad (45)$$

$$y_j^{t+1} = y_j^t - \lambda f_a^T(\zeta_1^t, \zeta_2^t)(y_i^t - y_j^t), \quad (46)$$

$$y_k^{t+1} = y_k^t - \lambda f_r^T(\zeta_1^t, \zeta_2^t)(y_i^t - y_k^t), \quad (47)$$

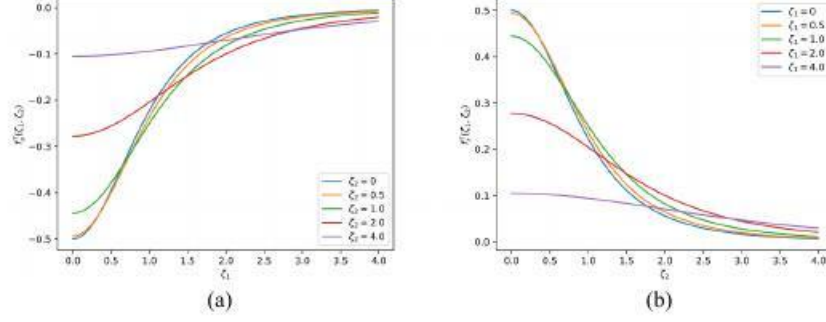


Figure 12: (a) Attraction shapes of TriMap across differing ζ_2 , and (b) repulsion shapes of TriMap across differing ζ_1 .

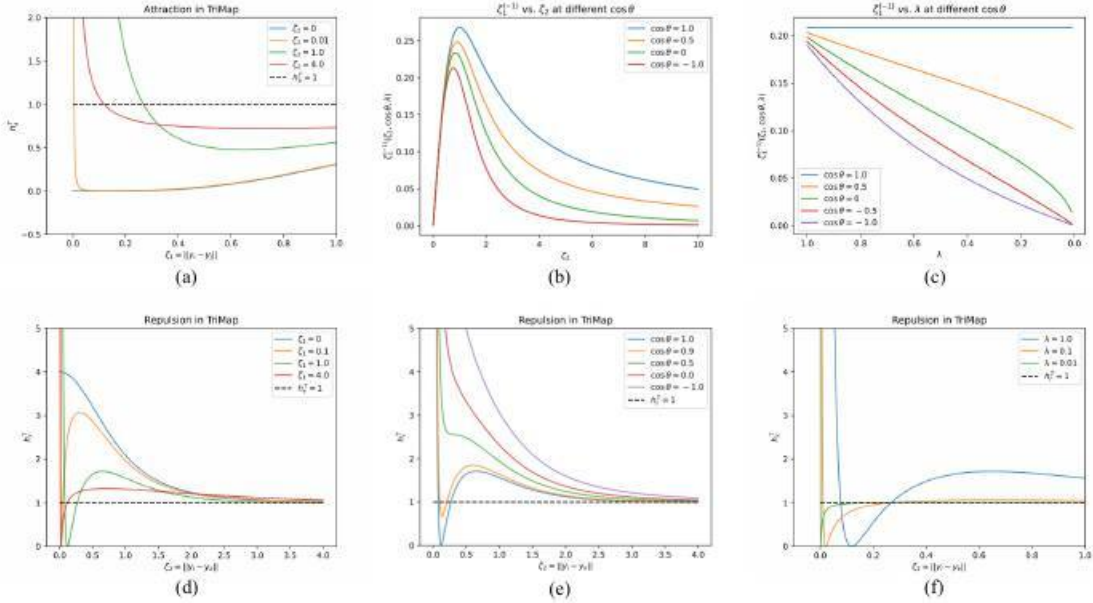


Figure 13: Attraction and repulsion behavior in TriMap. (a) h_a^T versus ζ_1 for differing ζ_2 ; values below the dotted line betoken attraction. As with UMAP, TriMap exhibits both attraction and repulsion for nearest neighbors (y_i, y_j). (b,c) In contradistinction to UMAP, the minimum distance for contraction ($\zeta_1^{(-1)}$) varies by virtue of its dependence upon (b) ζ_2 and (c) λ , the function $\cos \theta$ regulating the range of these values. (d-f) Repulsion in TriMap upon varying (d) ζ_1 , (e) $\cos \theta$, and (f) λ . Values above (below) the dotted line betoken repulsion (attraction). Whereas the repulsive force of UMAP yields repulsion alone, that of TriMap can furnish both attraction and repulsion. Unless otherwise labeled, $\zeta_1 = 1.0$, $\zeta_2 = 0.5$, $\cos \theta = 1.0$, and $\lambda = 1.0$.

where $\zeta_1 = \|y_i - y_j\|_2$ denotes the distance between nearest neighbors, $\zeta_2 = \|y_i - y_k\|_2$ the distance from the distant point, and f_a^T and f_r^T the attraction and repulsion shapes of TriMap, respectively. In contradistinction to UMAP, the attractive and repulsive components are non-separable and the shapes are contingent upon two distance measures (whence their two-dimensional character). The functional form of the attraction shape is

$$f_a^T(\zeta_1, \zeta_2) = -\frac{2(1 + \zeta_2^2)}{(2 + \zeta_1^2 + \zeta_2^2)^2}, \quad (48)$$

and the repulsion shape is

$$f_r^T(\zeta_1, \zeta_2) = \frac{2(1 + \zeta_1^2)}{(2 + \zeta_1^2 + \zeta_2^2)^2}. \quad (49)$$

The attraction and repulsion shapes of TriMap (Fig. 12) exhibit trends akin to those of UMAP. The minimum value of the repulsion shape is, however, -0.5 ; hence, unlike UMAP, TriMAP admits of no position flipping on account of attraction alone. Yet, inasmuch as Eqs. (45)-(47) are not decoupled as between attractive and repulsive terms, Propositions 4.1 and 4.2 do not apply. Addressing attraction first, we show

Proposition G.1. *Update equations (45)-(47) provide a contraction if*

$$h_a^T(\zeta_1, \zeta_2, \theta, \lambda) < 1, \quad (50)$$

where $h_a^T(\zeta_1, \zeta_2, \theta, \lambda) = (1 + 2\lambda f_a^T)^2 + 2(1 + 2\lambda f_a^T)\lambda f_r^T \frac{\zeta_2}{\zeta_1} \cos \theta + (\lambda f_r^T)^2 \frac{\zeta_2^2}{\zeta_1^2}$ and θ is the angle between the vectors $(y_i - y_j)$ and $(y_i - y_k)$, i.e., $\cos \theta = \frac{(y_i^t - y_j^t)^T (y_i^t - y_k^t)}{\|y_i^t - y_j^t\|_2 \|y_i^t - y_k^t\|_2}$.

Proof. We require

$$\|y_i^{t+1} - y_j^{t+1}\|_2^2 < \|y_i^t - y_j^t\|_2^2. \quad (51)$$

From Eq. (45) and (46): $y_i^{t+1} - y_j^{t+1} = (1 + 2\lambda f_a^T)(y_i^t - y_j^t) + \lambda f_r^T(y_i^t - y_k^t)$. Putting this value in Eq. (51), we obtain the desired inequality. $\square$

Inequality (50) depends upon ζ_1 , ζ_2 , $\cos \theta$, and λ . In particular, the value of $\zeta_1 = \|y_i - y_j\|_2$ —precisely where we desire a contraction—is coupled with the remaining variables. Figure 13(a) charts the attraction behavior for various values of ζ_2 , with $\cos \theta = 1$ and $\lambda = 1$. Values below the dotted line betoken attraction (and hence a contraction of the distance ζ_1), whereas values above it betoken repulsion (and hence an expansion of ζ_1). The point at which the dotted line and h_a^T intersect furnishes the minimum distance for contraction ($\zeta_1^{(-1)}$) [analogous to ζ_{-1} of UMAP], which we define as

$$\zeta_1^{(-1)}(\zeta_2, \theta, \lambda) = \arg \min_{\zeta_1} |h_a^T(\zeta_1, \zeta_2, \theta, \lambda) - 1|, \quad (52)$$

$$\text{s.t. } \zeta_1 \geq 0. \quad (53)$$

$\zeta_1^{(-1)}$ assumes a finite value and is > 0 in the majority of cases (Fig. 13(b,c)). Consequently, TriMap may display behavior analogous to that of UMAP and thus require learning-rate annealing or some kindred device (in the TriMap implementation, the authors resort to the delta-bar-delta (Jacobs, 1988) method under an appropriate initialization).

Proposition G.2. *Update equations (45)-(47) provide expansion if*

$$h_r^T(\zeta_1, \zeta_2, \theta, \lambda) > 1, \quad (54)$$

where $h_r^T(\zeta_1, \zeta_2, \theta, \lambda) = (1 + 2\lambda f_r^T)^2 + 2\lambda f_a^T(1 + \lambda f_r^T) \frac{\zeta_1}{\zeta_2} \cos \theta + (\lambda f_a^T)^2 \frac{\zeta_1^2}{\zeta_2^2}$.

Proof. We require

$$\|y_i^{t+1} - y_k^{t+1}\|_2^2 > \|y_i^t - y_k^t\|_2^2. \quad (55)$$

From Eq. (45) and (47): $y_i^{t+1} - y_k^{t+1} = (1 + 2\lambda f_r^T)(y_i^t - y_k^t) + \lambda f_a^T(y_i^t - y_j^t)$. Putting this value in Eq. (55) we obtain inequality (54). $\square$

Inequality (54) likewise depends upon the set ζ_1 , ζ_2 , $\cos \theta$, and λ . Here our interest lies in the expansion of $\zeta_2 = \|y_i - y_k\|_2$. Figures 13(d-f) chart the repulsion behavior as the remaining quantities are varied. Values above the dotted line betoken repulsion (and thus an expansion of distance), while those below it betoken attraction. The striking divergence from UMAP is that repulsion in TriMap can occasion contraction in lieu of expansion; since this anomaly manifests only at small distances, it can be circumvented through an appropriate initialization and a judicious choice of triplets.

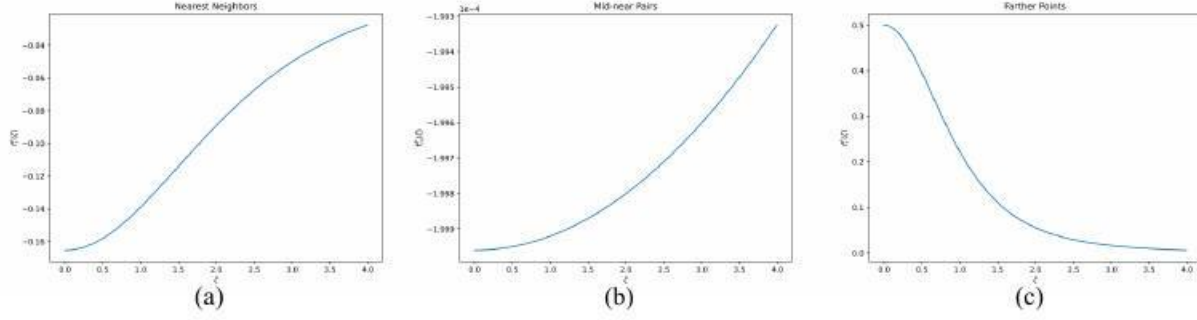


Figure 14: Attraction and repulsion shapes of PaCMAP. (a,b) Attraction shapes for the (a) nearest-neighbor and (b) mid-near points (note that the values are on the order of 10^{-4}). (c) Repulsion shape for the farther pairs. Here $\lambda = 1$ throughout all panels.

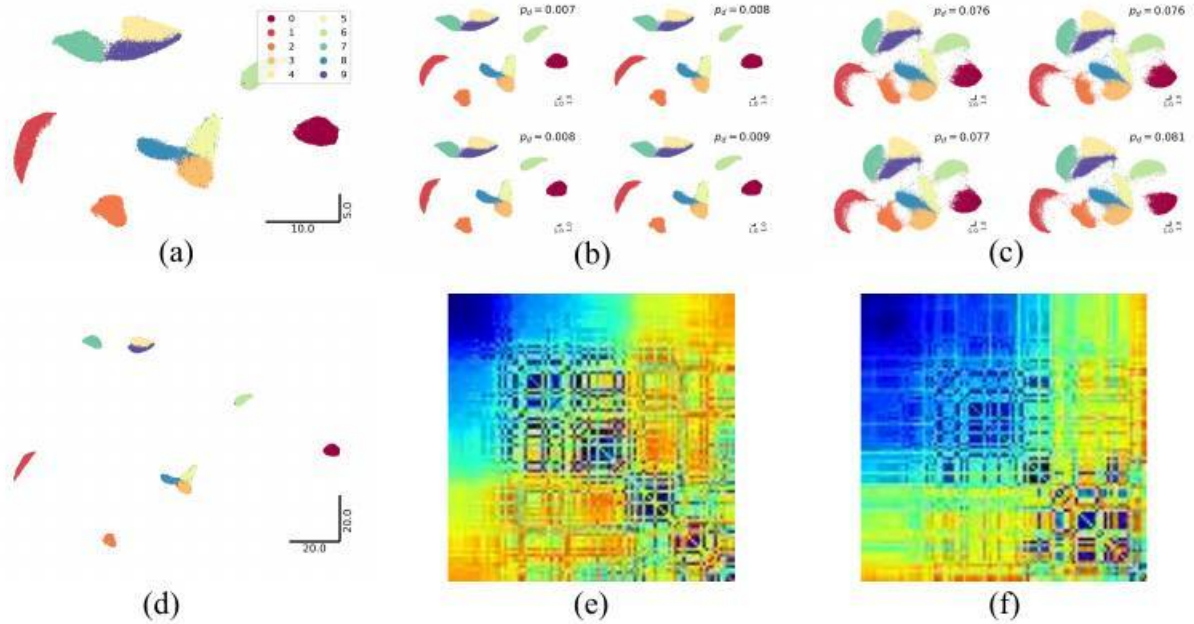


Figure 15: PaCMAP behavior under differing conditions. (a) PaCMAP of the MNIST data with PCA initialization. (b,c) The four samples that best match (a), obtained by (b) the original PaCMAP and (c) PaCMAP with a modified attraction shape, $f_a^M(\zeta) = f_a^P(\zeta) - 0.001\zeta$, when randomly initialized. (d) PaCMAP of MNIST with a modified repulsion shape $f_r^M = f_r^P + 0.00005$. (e,f) Procrustes matrices for (e) the original PaCMAP (0.51 ± 0.22) and (f) PaCMAP with a modified attraction shape (0.43 ± 0.22). As with UMAP, heightened attraction at farther distances yields improved consistency, whereas heightened repulsion yields smaller clusters and larger inter-cluster distances.

G.2 Pairwise Controlled Manifold Approximation (PaCMAP)

PaCMAP (Wang et al., 2021) optimizes the low-dimensional embedding across a range of scales. Its loss function is

$$\mathcal{L}^P = w_{NB} \sum_{(i,j) \in NN} \frac{1}{1 + 10q_{ij}} + w_{MN} \sum_{(i,k) \in MN} \frac{1}{1 + 10000q_{ik}} + w_{FP} \sum_{(i,l) \in FP} \frac{1}{1 + \frac{1}{q_{il}}}, \quad (56)$$

where w_{NB} , w_{MN} , and w_{FP} denote the weights of the nearest-neighbor (NN) pairs, mid-near (MN) pairs, and further pairs (FP), respectively (the details appear in Appendix H). The first two terms furnish attraction, whereas the last furnishes repulsion. A closer scrutiny of the loss function reveals it to be a modified form of TriMap’s triplet loss: for the attractive terms, it supplants TriMap’s affinity for distant points with constant terms ($1/10$ for nearest neighbors, $1/10000$ for mid-near pairs, and 1 for farther points). The loss function is thereby separable into three terms and

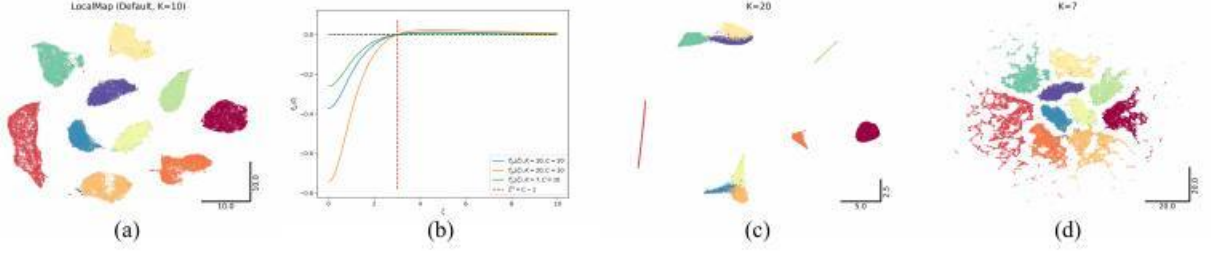


Figure 16: The behavior of LocalMAP on the MNIST data. (a) The default embedding. (b) The attraction–repulsion shape of the nearest neighbors; a value below (above) the dotted line betokens attraction (repulsion). The transition from attraction to repulsion supervenes as ζ grows and crosses $\sqrt{C} - 1$. Following the LocalMAP implementation, we adopt $C = 10$. (c) When K is large ($= 20$), repulsion prevails and the clusters grow compact. (d) When K is small ($= 7$), attraction prevails and the clusters fragment.

decouples the update equations. The update equations of the nearest-neighbor term are

$$y_i^{t+1} = y_i^t + \lambda f_a^P(\zeta_1^t)(y_i^t - y_j^t), \quad (57)$$

$$y_j^{t+1} = y_j^t - \lambda f_a^P(\zeta_1^t)(y_i^t - y_j^t), \quad (58)$$

of the mid-near pairs are

$$y_i^{t+1} = y_i^t + \lambda f_m^P(\zeta_2^t)(y_i^t - y_k^t), \quad (59)$$

$$y_k^{t+1} = y_k^t - \lambda f_m^P(\zeta_2^t)(y_i^t - y_k^t), \quad (60)$$

and of the farthest pairs are

$$y_i^{t+1} = y_i^t + \lambda f_r^P(\zeta_3^t)(y_i^t - y_l^t), \quad (61)$$

$$y_l^{t+1} = y_l^t - \lambda f_r^P(\zeta_3^t)(y_i^t - y_l^t), \quad (62)$$

where $\zeta_1 = \|y_i - y_j\|_2$, $\zeta_2 = \|y_i - y_k\|_2$, and $\zeta_3 = \|y_i - y_l\|_2$ denote the respective distances, f_a^P and f_m^P the attraction shapes for the nearest-neighbor and mid-near pairs, respectively, and f_r^P the repulsion shape for the farthest pairs. The corresponding functional forms of the shapes are

$$f_a^P(\zeta) = -\frac{20}{(11 + \zeta^2)^2}, \quad (63)$$

$$f_m^P(\zeta) = -\frac{20000}{(10001 + \zeta^2)^2}, \quad (64)$$

$$f_r^P(\zeta) = \frac{2}{(2 + \zeta^2)^2}. \quad (65)$$

f_a^P and f_m^P conform to Proposition 4.1 and f_r to Proposition 4.2 (Fig. 14). The attraction is quite modest by comparison with UMAP, yet it proves adequate across a broad range of learning rates (modulated by the Adam algorithm (Kingma and Ba 2015)); with $w_{NB} = 3$, the maximum attraction remains everywhere below 0.5, thereby forestalling any flips during the attractive update. PaCMap customarily initializes the embedding within a small sphere in the low dimension (in (Wang et al. 2021), for instance, the initialization is frequently on the order of 10^{-3}) and relies upon repulsion to separate the individual clusters. On the whole, it largely recovers UMAP’s clustering properties—especially for MNIST—with an improved ordering. Its consistency under random initialization surpasses that of UMAP (Fig. 15 b,e) and may be improved yet further through a modified attraction (Fig. 15 c,f). Intensifying repulsion by adding a small value to the repulsion shape augments the compactness of the embedding (by enlarging the inter-cluster distance).

G.3 Pairwise Controlled Manifold Approximation with Local Adjusted Graph (LocalMAP)

LocalMAP (Wang et al. 2025) constitutes a further iteration of the PaCMap algorithm. Among its defining features is the separation of all 10 clusters of the MNIST data (Fig. 16 a), a behavior kindred to that displayed in Figs. 3 g,h,j,k) and Figs. 4 a,b). Here we explore the interplay of the attractive and repulsive forces as it bears upon the compactness and connectedness of clusters.

LocalMAP first executes PaCMAP and thereafter conducts additional optimization upon the attraction so as to decouple certain clusters. To this end it minimizes the following loss function

$$\mathcal{L}^L = \sum_{(i,j) \in NN} \frac{K}{\frac{1}{\sqrt{q_{ij}}} + C\sqrt{q_{ij}}} + \sum_{(i,l) \in FP} \frac{1}{1 + \frac{1}{q_{il}}}. \quad (66)$$

The first term amalgamates the attractive and repulsive character of the triplet loss function operating upon a single pair: in one regime this function occasions attraction, in the other repulsion. The second term is identical to those of PaCMAP, the sole difference being that the algorithm resamples the further pairs every few iterations. We therefore confine our analysis to the first term, involving the nearest neighbors. The update equations are

$$y_i^{t+1} = y_i^t + \lambda f_{ar}^L(\zeta_1^t)(y_i^t - y_j^t), \quad (67)$$

$$y_j^{t+1} = y_j^t - \lambda f_{ar}^L(\zeta_1^t)(y_i^t - y_j^t), \quad (68)$$

where f_{ar}^L is the attraction-repulsion shape given by

$$f_{ar}^L(\zeta) = -\frac{K(C-1-\zeta^2)}{2\sqrt{1+\zeta^2}(1+C+\zeta^2)^2}. \quad (69)$$

The update effects contraction so long as $\zeta^2 < C-1$ and $-1 < \lambda f_{ar}^L < 0$ (from Proposition 4.1). When $\zeta^2 > C-1$, we have $f_{ar} > 0$, and by Proposition 4.2 the update equations effect expansion. The values of K and C dictate whether attraction or repulsion predominates in the dynamics. In the LocalMAP implementation, both C and K are set to 10 (Fig. 16b)), and the strength of attraction exceeds that of PaCMAP (Fig. 14a)). Consequently, when $\zeta > 3$, the nearest neighbors encounter repulsion, whereupon pairs bridging two clusters are driven apart; the value of K , functioning as a scaling parameter for the forces, regulates this separation.

By Proposition 4.1 the condition $\lambda f_{ar}^L(0) \geq -\frac{1}{2}$ furnishes the maximum values of K —and the maximum attraction attainable by f_{ar}^L —without flipping the placements of the pairs. (It is desirable to avert such flipping during the LocalMAP optimization, so as to preserve the ordering inherited from PaCMAP; were it otherwise, cluster separation might be impeded.) This simplifies (69) to $K \leq \frac{\lambda(1+C)^2}{C-1}$, which for $C = 10$ and $\lambda = 1$ yields $K \lesssim 13.44$. Moreover, at a higher value of $K \gtrsim 13.44$, the repulsive forces predominate and the clusters grow more compact, yet the objective of LocalMAP miscarries (the bridge between clusters persists in Fig. 16c) for $K = 20$). As K decreases, conversely, the attractive forces predominate (the repulsive forces being too feeble), and the embedding evinces a fragmentation of the extant clusters (Fig. 16d) for $K = 7$, which echoes that of Fig. 4b)).

G.4 t -distributed Stochastic Neighbor Embedding (t -SNE)

t -SNE (Van der Maaten and Hinton 2008) optimizes pairwise distances. Its loss function is

$$\mathcal{L} = -\sum_{i,j} w_{i,j} \log \left(\frac{q_{i,j}}{\sum_{k \neq l} q_{k,l}} \right), \quad (70)$$

where $w_{i,j}$ denotes the weight of the pair. The original implementation of t -SNE takes account of all pairs, and not merely the nearest neighbors. This loss function decomposes into attractive and repulsive forces as

$$\mathcal{L} = \sum_{i,j} \left[-w_{i,j} \log(q_{i,j}) + w_{i,j} \log \left(\sum_{k \neq l} q_{k,l} \right) \right]. \quad (71)$$

As before, the first term furnishes the attractive forces and the second the repulsive forces. While the attractive term is identical to that of UMAP (with $a = 1$ and $b = 1$) and is readily computed, the repulsive term is coupled across every pair and is, in consequence, exceedingly costly. Invoking the same principles as we applied to UMAP, we may set down the update equations of t -SNE. Since the original t -SNE did not rely upon the nearest-neighbor graph, the weight $w_{i,j}$ —computed over all pairs—assumes a signal importance in the update equations. The attractive update equations are

$$y_i^{t+1} = y_i^t + \lambda w_{i,j} f_a^{t-SNE}(\zeta_{i,j}^t)(y_i^t - y_j^t), \quad (72)$$

$$y_j^{t+1} = y_j^t - \lambda w_{i,j} f_a^{t-SNE}(\zeta_{i,j}^t)(y_i^t - y_j^t), \quad (73)$$

where f_a^{t-SNE} denotes the attraction shape of t -SNE and $\zeta_{i,j} = \|y_i - y_j\|_2$. The update equations for the repulsive components are

$$y_i^{t+1} = y_i^t + \lambda \frac{w_{i,j}}{Z} \sum_k f_r^{t-SNE}(\zeta_{i,k}^t)(y_i^t - y_k^t), \quad (74)$$

$$y_j^{t+1} = y_j^t - \lambda \frac{w_{i,j}}{Z} \sum_l f_r^{t-SNE}(\zeta_{l,j}^t)(y_l^t - y_j^t), \forall j, j \neq i, \quad (75)$$

where f_r^{t-SNE} denotes the repulsion shape of t -SNE and $Z = \sum_{k \neq l} q_{k,l}$. The functional forms of these shapes are

$$f_a^{t-SNE}(\zeta) = -\frac{2}{1 + \zeta^2}, \text{ and} \quad (76)$$

$$f_r^{t-SNE}(\zeta) = \frac{2}{(1 + \zeta^2)^2}. \quad (77)$$

From the attractive update Eqs. (72)-(73), f_a^{t-SNE} conforms to Proposition 4.1 (with $0 < \lambda w_{i,j} f_a^{t-SNE} < -1$) and furnishes the minimum distance for contraction, ζ_{-1} . The repulsive update, being coupled across all pairs, bears no simple relation to the repulsion shape; drawing instead upon the experience garnered from our analysis of TriMap, we may derive the following:

Proposition G.3. The update Eqs. (74)-(75) provide an expansion if

$$h_r^{t-SNE}(\zeta_{i,j}, v, \theta, \lambda, w_{i,j}) > 1, \quad (78)$$

where

$$h_r^{t-SNE}(\zeta_{i,j}, v, \theta, \lambda, w_{i,j}) = (1 + 2\lambda \frac{w_{i,j}}{Z} f_r^{t-SNE}(\zeta_{i,j}))^2 + \frac{\|v\|_2^2}{\zeta_{i,j}^2} + 2\lambda \frac{w_{i,j}}{Z} f_r^{t-SNE}(\zeta_{i,j}) \frac{\|v\|_2}{\zeta_{i,j}} \cos \theta,$$

$$v = \lambda \frac{w_{i,j}}{Z} \left(\sum_{k, k \neq j} f_r^{t-SNE}(\zeta_{i,k})(y_i - y_k) + \sum_{l, l \neq i} f_r^{t-SNE}(\zeta_{l,j})(y_l - y_j) \right),$$

and θ is the angle between $(y_i - y_j)$ and v .

Proof. We require,

$$\|y_i^{t+1} - y_j^{t+1}\|_2^2 > \|y_i^t - y_j^t\|_2^2. \quad (79)$$

From Eqs. (74) and (75), $y_i - y_j = (1 + 2\lambda \frac{w_{i,j}}{Z} f_r^{t-SNE}(\zeta_{i,j}))(y_i - y_j) + v$. Putting this value in Eq. (79), we obtain the desired inequality. $\square$

Inasmuch as t -SNE's condition for repulsion (Eq. 78) resembles that of TriMap, the repulsion behavior will be identical; t -SNE's repulsive forces can accordingly furnish both attraction and repulsion.

Since t -SNE's forces are scaled (by $w_{i,j}$ for attraction and by $\frac{w_{i,j}}{Z}$ for repulsion), both the attractive and repulsive forces are typically weaker than those of UMAP. In consequence, the algorithm generally employs large values of the learning rate (for example, $\approx 10^3$ in the Open- t -SNE package (Poličar et al. 2024)). Most t -SNE implementations, moreover, require an 'early exaggeration' step, wherein the attractive forces are multiplied by a constant value over the first few iterations. This impels points that ought to be proximate but are presently situated far apart to draw together (thereby inducing far-sightedness); conversely, should certain points lie very close ($\zeta_{i,j} \approx \zeta_{-1}$) yet require separation, the early-exaggeration trick accomplishes this as well. 'Early exaggeration' thus discharges a vital office in securing a consistent embedding in t -SNE and constitutes an indispensable feature of the algorithm, most especially under random initialization. (This same trick, when applied throughout the optimization, renders t -SNE embeddings more nearly akin to those of UMAP (Böhm et al. 2022).)

G.5 Stochastic Neighbor Embedding (SNE)

SNE (Hinton and Roweis 2002) numbers among the oldest algorithms of this class. It abides by the same formula as t -SNE (Eq. 71), save that $q_{i,j} = \exp(-\|y_i - y_j\|_2^2)$. This gives rise to the attraction shape

$$f_a^{SNE}(\zeta) = -2, \quad (80)$$

that follows Proposition 4.1 (with $0 < \lambda w_{i,j} f_a^{SNE} < -1$) and the repulsion shape

$$f_r^{SNE}(\zeta) = 2 \exp(-\zeta^2), \quad (81)$$

that conforms to Proposition G.3 (upon substituting the SNE counterparts for the t -SNE symbols). The attraction shape is ill-posed and therefore depends chiefly upon the values of the learning rate (λ) and the weights ($w_{i,j}$) to effect contraction, whence clusters overlap one another even for a modest number of samples (the so-called crowding problem (Van der Maaten and Hinton 2008))—a deficiency that t -SNE and its successors remedy by forsaking the Gaussian kernel in favor of a heavy-tailed one, namely Eq. (43).

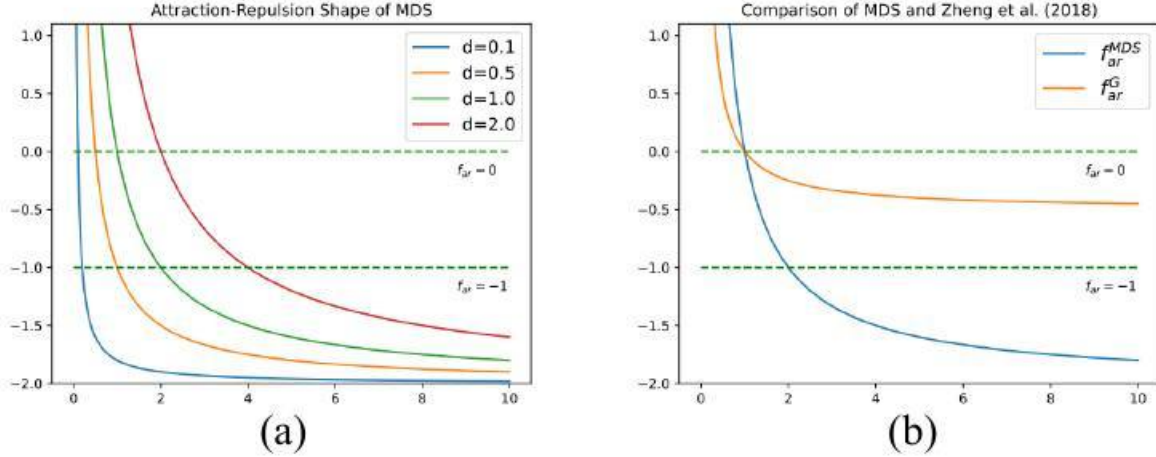


Figure 17: (a) The attraction–repulsion characteristic of MDS as expressed through f_{ar} . (b) A comparison of the attraction–repulsion characteristics of MDS and of the method in (Zheng et al. 2018) for $d = 1.0$.

G.6 Multidimensional Scaling (MDS)

Multidimensional scaling (MDS) (Borg and Groenen, 2007) does not, as a rule, resort to gradient methods—these often failing to converge upon satisfactory mappings—but rather employs stress majorization. A few works nonetheless treat of gradient methods (Kruskal, 1964; Zheng et al. 2018), and we offer a brief treatment of the matter below. In particular, Zheng et al. (2018) formulates a successful gradient-descent-based MDS algorithm for graph drawing. We begin with the cost function (we may disregard the weight $w_{i,j}$ without loss of generality to their approach, subsuming it, for the purposes of discussion, into the learning rate):

$$\mathcal{L}^{MDS} = \sum_{i,j,i < j} (\|y_i - y_j\| - d_{ij})^2. \quad (82)$$

The loss comprises a singleton term for each pair and possesses no explicit attractive and repulsive terms; the term itself furnishes both attraction and repulsion. Fortunately, the loss is separable into its individual (i, j) components, which permits us to analyze a single pair. The derivative is given by

$$\frac{\partial}{\partial y_i} (\|y_i - y_j\| - d_{ij})^2 = 2 \frac{\|y_i - y_j\| - d_{ij}}{\|y_i - y_j\|} (y_i - y_j). \quad (83)$$

The attraction–repulsion shape of MDS thus becomes

$$f_{ar}^{MDS}(\zeta) = -2 \frac{\zeta - d}{\zeta}, \quad (84)$$

where $\zeta = \|y_i - y_j\|_2$. Here the shape is attractive when $\zeta > d$, and within this range it is bounded within $[-2, 0]$ (Fig. 17(a)). This, however, is ill-suited to convergence, since only values within $[-1, 0]$ contract while the remainder expand; and, to compound the difficulty, values below -2 cause the points to flip and expand. To circumvent this, Zheng et al. (2018) adopts an ad-hoc formulation inspired by force-directed graphs, given by

$$f_{ar}^G(\zeta) = -\frac{1}{2} \frac{\zeta - d}{\zeta}, \quad (85)$$

which operates for an effective learning rate $\lambda \leq 1$. Since f_{ar}^G is bounded within $[-0.5, 0]$, $\lambda \leq 1$ indeed suffices. For $\zeta < d$, on the other hand, $f_{ar}^{MDS} (> 0)$ exhibits repulsive behavior. On the whole, this attraction–repulsion interaction operates to best effect when the initialization already lies close to the desired output. Should one persist in optimizing a given pair, it will oscillate about the distance ($\zeta = d$) at which $f_{ar}^G = 0$ (whence we may frame a notion of distance analogous to ζ_{-1} of UMAP). No choice of learning rate reduces this distance to zero (this, in fact, is precisely the objective of MDS). The clusters are therefore frequently fuzzy by comparison with methods such as UMAP and PaCMAP (for pertinent illustrations, see Lambert et al. (2022), de Bodt et al. (2025), and Kury et al. (2025)). It bears noting that Lambert et al. (2022) converts the MDS loss function into a quartet stress (a loss involving four samples) together with a relative distance (the distance divided by the sum of the six distances in the quartet), thereby enabling global regularization and a computation more economical than the majorization approach.

H Construction of the High-Dimensional Graph

Stochastic Neighbor Embedding (SNE) [\[Hinton and Roweis, 2002\]](#) furnishes the foundation upon which modern dimensionality reduction algorithms rest. It constructs a high-dimensional graph of the dataset $X = \{x_i \in R^n | i = 1, \dots, N\}$ by way of the following system of equations:

$$w_{ij} = \frac{p_{j|i} + p_{i|j}}{2N}, \quad (86)$$

$$w_{j|i} = \frac{\exp(-\|x_i - x_j\|_2^2 / 2\sigma_i^2)}{\sum_{t \neq v} \exp(-\|x_t - x_v\|_2^2 / 2\sigma_t^2)}, \quad (87)$$

where σ_i^2 is so chosen as to accord with a user-defined value, the *perplexity* P , defined as

$$P = 2^{H_i} \quad (88)$$

$$H_i = \sum_{j \neq i} w_{j|i} \log_2 w_{j|i}. \quad (89)$$

UMAP, by contrast, constructs its high-dimensional graph through the following system of equations, which draws upon the k-nearest neighbor (k-NN) algorithm:

$$p_{i|j} = \begin{cases} \exp\left(-\frac{d(x_i, x_j) - \rho_i}{\sigma_i}\right) & \text{if } x_j \in \text{KNN}(x_i, k) \\ 0 & \text{otherwise} \end{cases}, \quad (90)$$

$$\rho_i = \min_{x_j \in \text{KNN}(x_i, k)} d(x_i, x_j), \quad (91)$$

where $\text{KNN}(x_i, k)$ denotes the set of k -nearest neighbors of the point x_i and σ_i a scaling parameter such that $\sum_j p_{i|j} = \log_2(k)$. The graph is thereafter symmetrized by means of a t-conorm:

$$p_{i,j} = p_{i|j} + p_{j|i} - p_{i|j}p_{j|i}. \quad (92)$$

PaCMAP employs the k-NN graph alone for the affinities (all set equal to 1), together with a self-tuning distance measure [\[Zelnik-Manor and Perona, 2004\]](#),

$$d_{i,j}^2 = \frac{\|x_i - x_j\|^2}{\sigma_i \sigma_j}, \quad (93)$$

where σ_i denotes the average distance between x_i and its fourth through sixth Euclidean nearest neighbors. Its purpose coincides with that of the corresponding σ_i parameters in Eqs. [\(87\)](#) and [\(90\)](#), the three being defined differently notwithstanding.

Whatever these choices, it is the optimization—and, by our analysis, the attraction and repulsion shapes—that constitute the primary drivers of the low-dimensional embedding.

I Detailed Results for Varying λ_a and λ_b

In this section we furnish detailed results for the experiments of Fig. 1(h,i). We obtained these results by transposing the attraction and repulsion shapes of UMAP into those of the other methods. Figure 18 reproduces the results reported in the main text. Figures 19, 22 display the individual embeddings for each choice of λ_a and λ_r across the several methods, together with their initialization, a reference embedding obtained when the learning rate λ is annealed, and the embeddings obtained when either λ_a or λ_r is set to 0. For PaCMAP, which invokes the notion of mid-near pairs, we adjoin a further reference in which mid-near pairs are included. Figures 23, 27 and Figures 28, 32 present the corresponding results on the FMNIST and single-cell transcriptomes datasets, respectively.

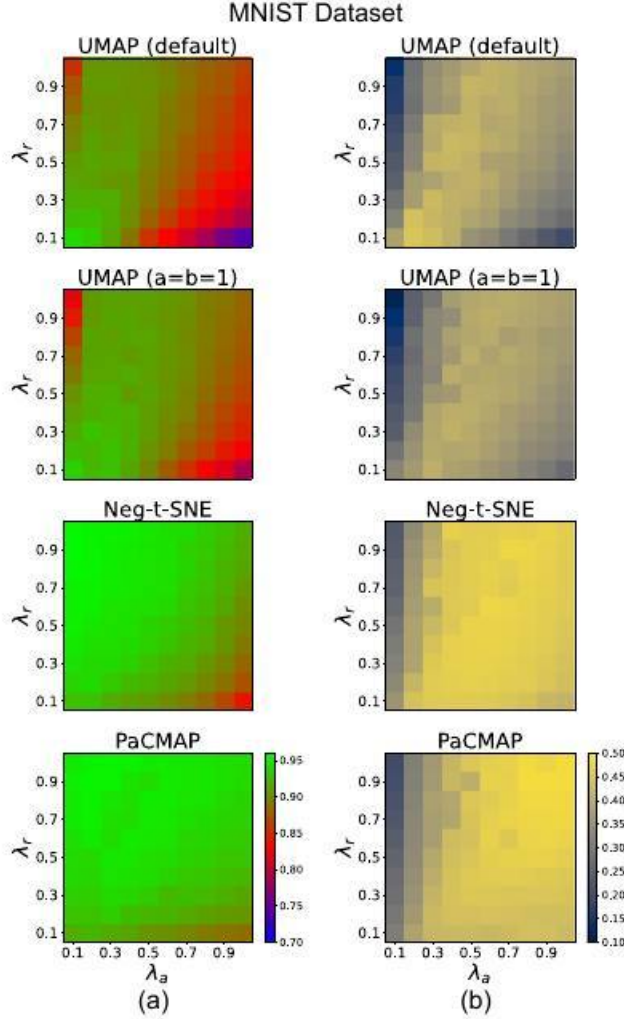


Figure 18: (a) The trustworthiness and (b) the silhouette score attained by the several methods on the MNIST dataset.

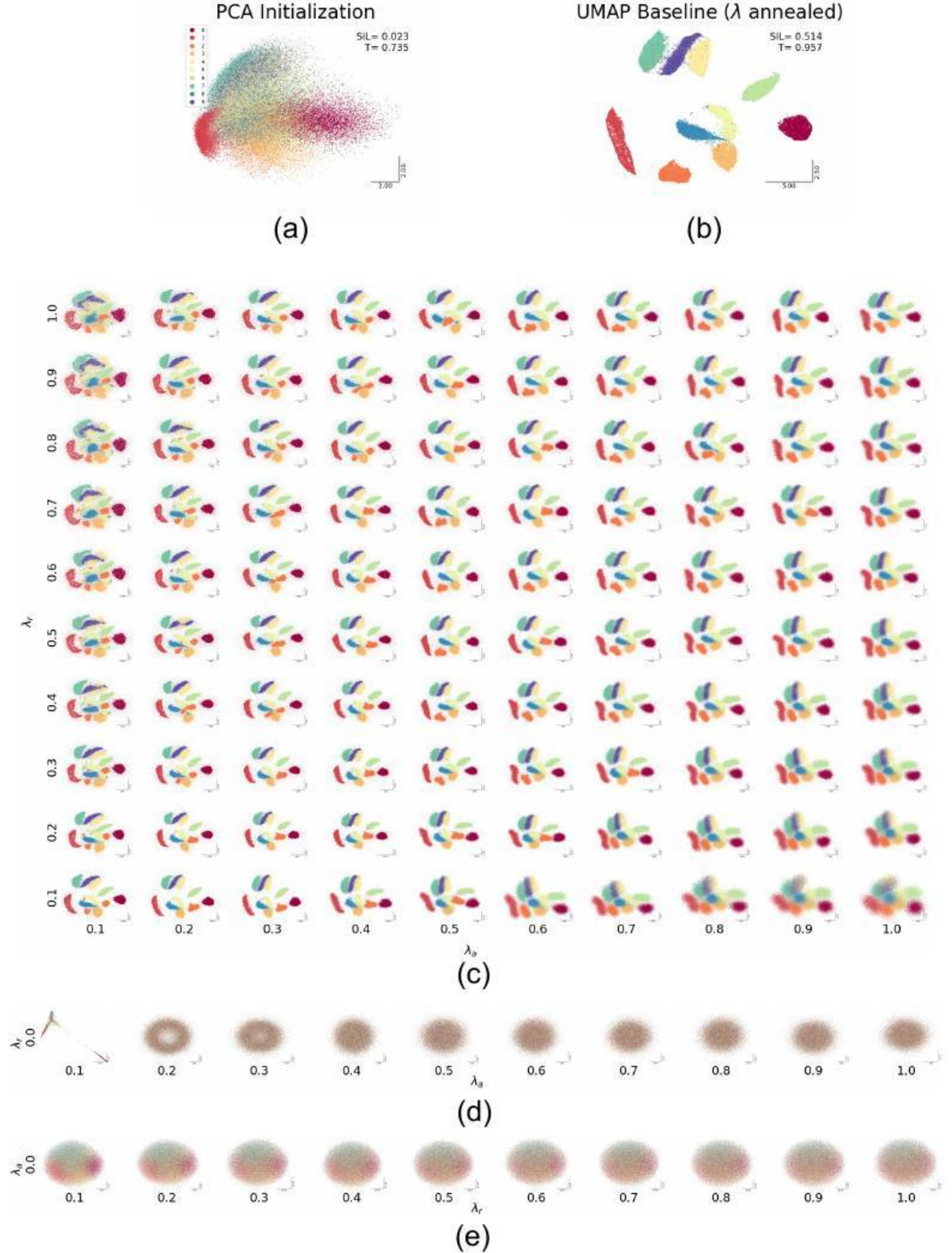


Figure 19: The variation of λ_a and λ_r for the MNIST dataset using UMAP’s attraction and repulsion shapes. (a) The initialization employed for the embeddings. (b) The baseline obtained when λ is annealed from 1. (c) The embeddings obtained as λ_a and λ_r are varied (without any annealing). (d) When λ_r is set to 0 (no repulsion), the attractive force alone proves unable to produce any cluster. (e) Likewise, when λ_a is set to 0 (no attraction), the repulsive force alone proves unable to produce any clusters.

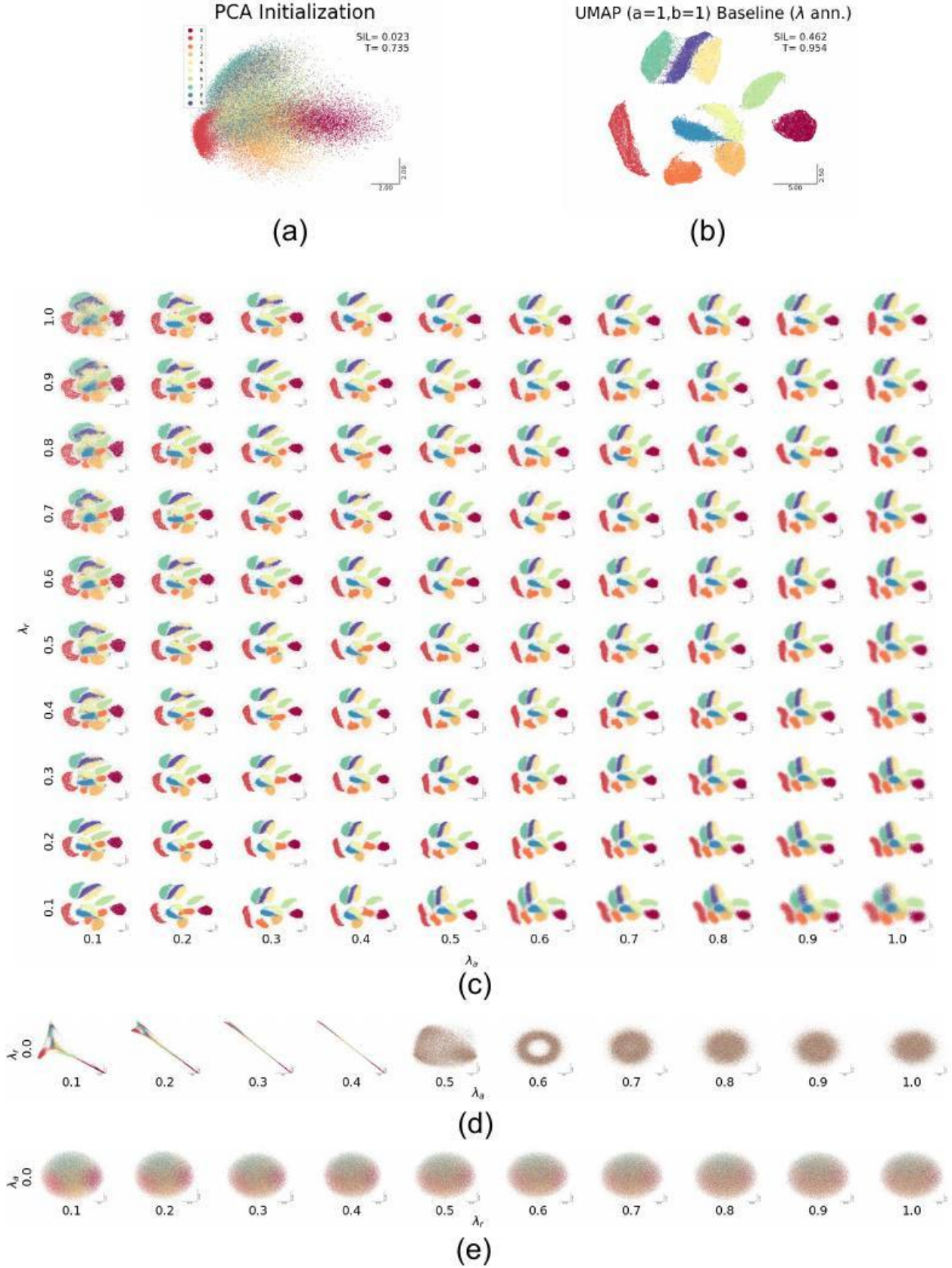


Figure 20: The variation of λ_a and λ_r for the MNIST dataset using UMAP's attraction and repulsion shapes (by setting $a = 1$ and $b = 1$). (a) The initialization employed for the embeddings. (b) The baseline obtained when λ is annealed from 1. (c) The embeddings obtained as λ_a and λ_r are varied (without any annealing). (d) When λ_r is set to 0 (no repulsion), the attractive force alone proves unable to produce any cluster. (e) Likewise, when λ_a is set to 0 (no attraction), the repulsive force alone proves unable to produce any clusters.

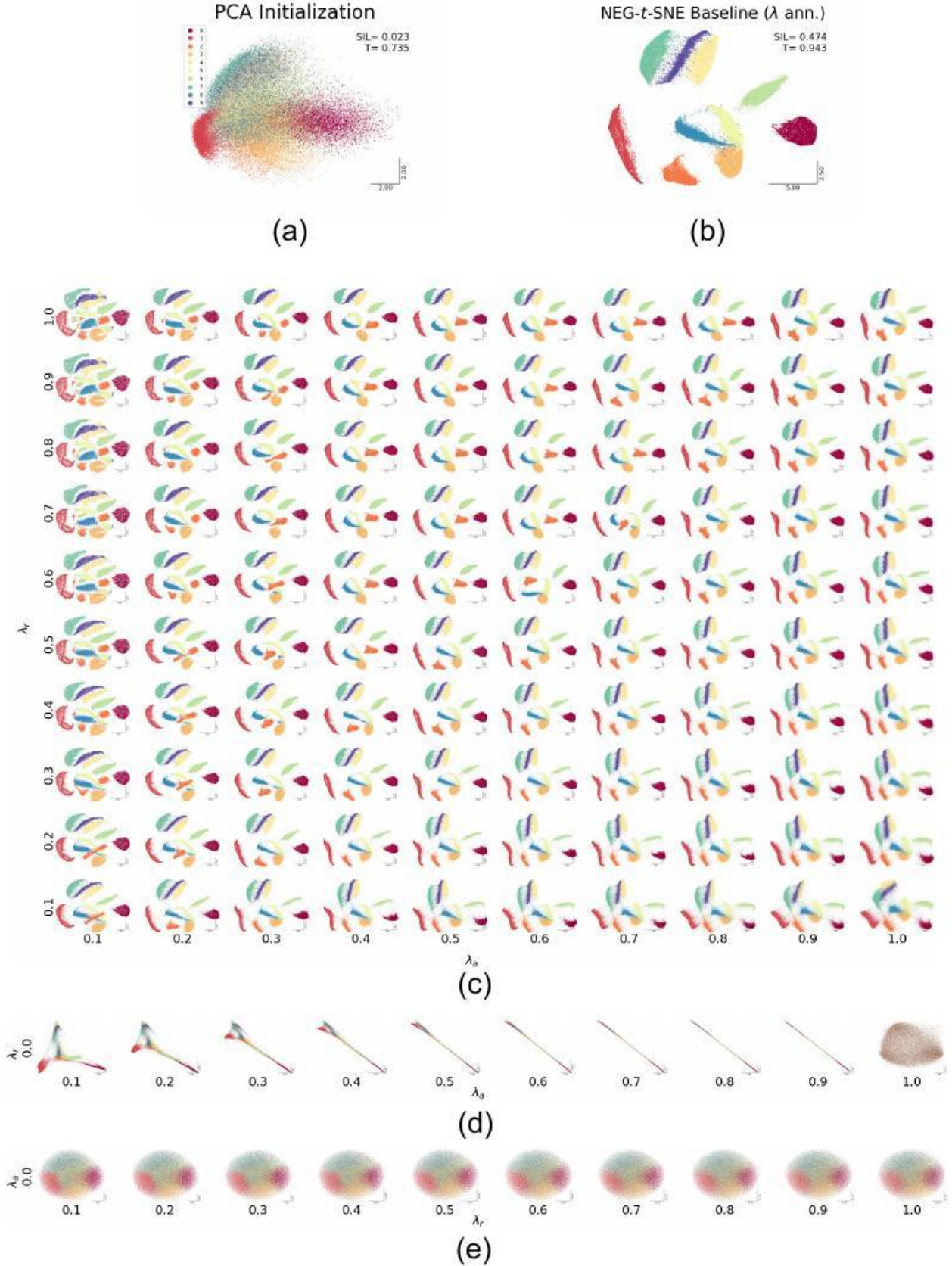


Figure 21: The variation of λ_a and λ_r for the MNIST dataset using NEG-t-SNE's attraction and repulsion shapes. (a) The initialization employed for the embeddings. (b) The baseline obtained when λ is annealed from 1. (c) The embeddings obtained as λ_a and λ_r are varied (without any annealing). (d) When λ_r is set to 0 (no repulsion), the attractive force alone proves unable to produce any cluster. (e) Likewise, when λ_a is set to 0 (no attraction), the repulsive force alone proves unable to produce any clusters.

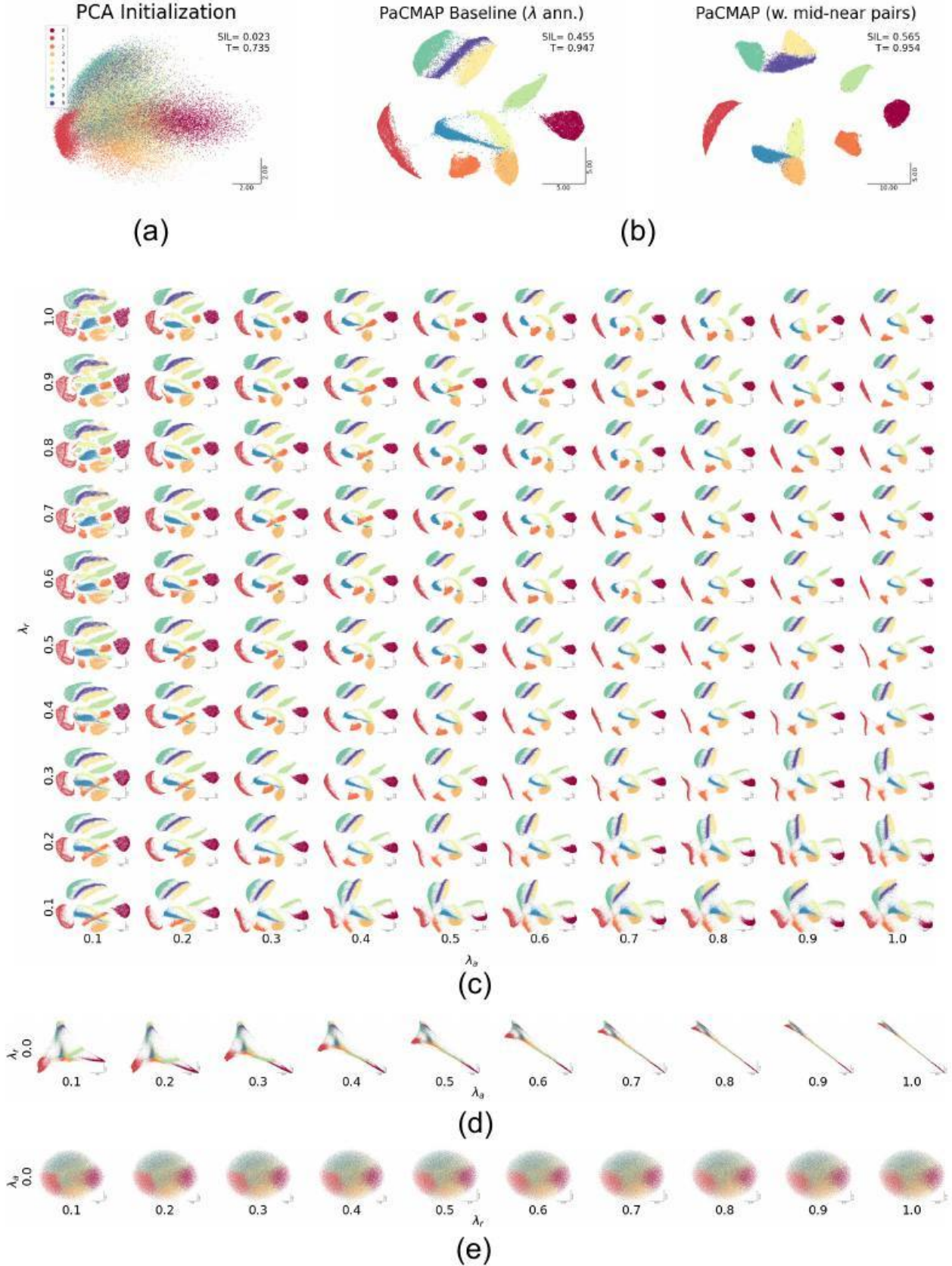


Figure 22: The variation of λ_a and λ_r for the MNIST dataset using PaCMAP’s attraction and repulsion shapes. (a) The initialization employed for the embeddings. (b) Left: the baseline obtained when λ is annealed from 1 (mid-near points being excluded so as to observe the attraction–repulsion interaction in isolation); right: the case in which mid-near points are taken into account. (c) The embeddings obtained as λ_a and λ_r are varied (without any annealing). (d) When λ_r is set to 0 (no repulsion), the attractive force alone proves unable to produce any cluster. (e) Likewise, when λ_a is set to 0 (no attraction), the repulsive force alone proves unable to produce any clusters.

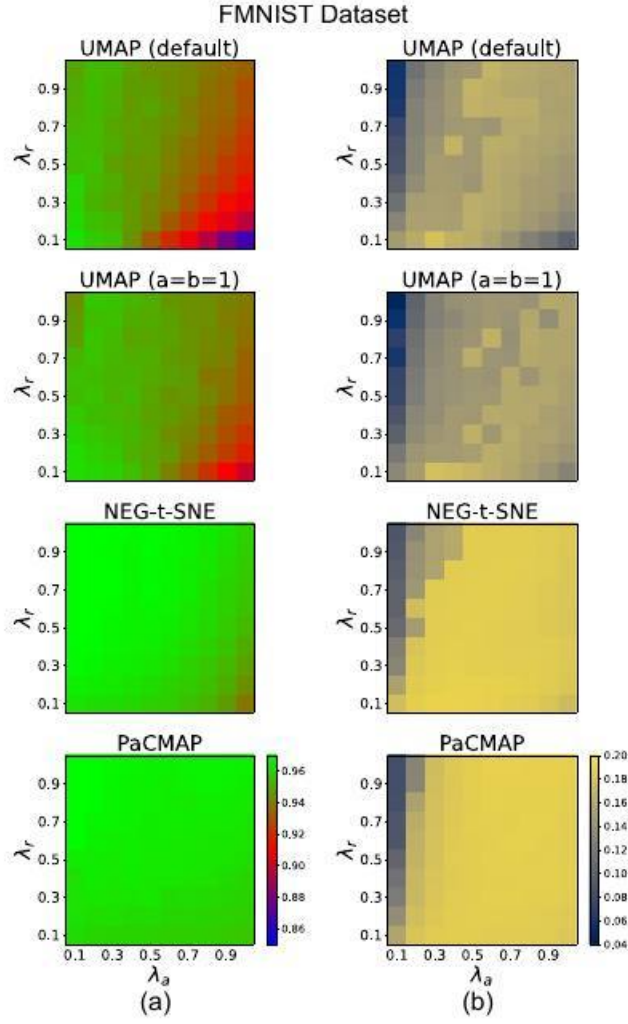


Figure 23: (a) The trustworthiness and (b) the silhouette score attained by the several methods on the FMNIST dataset.

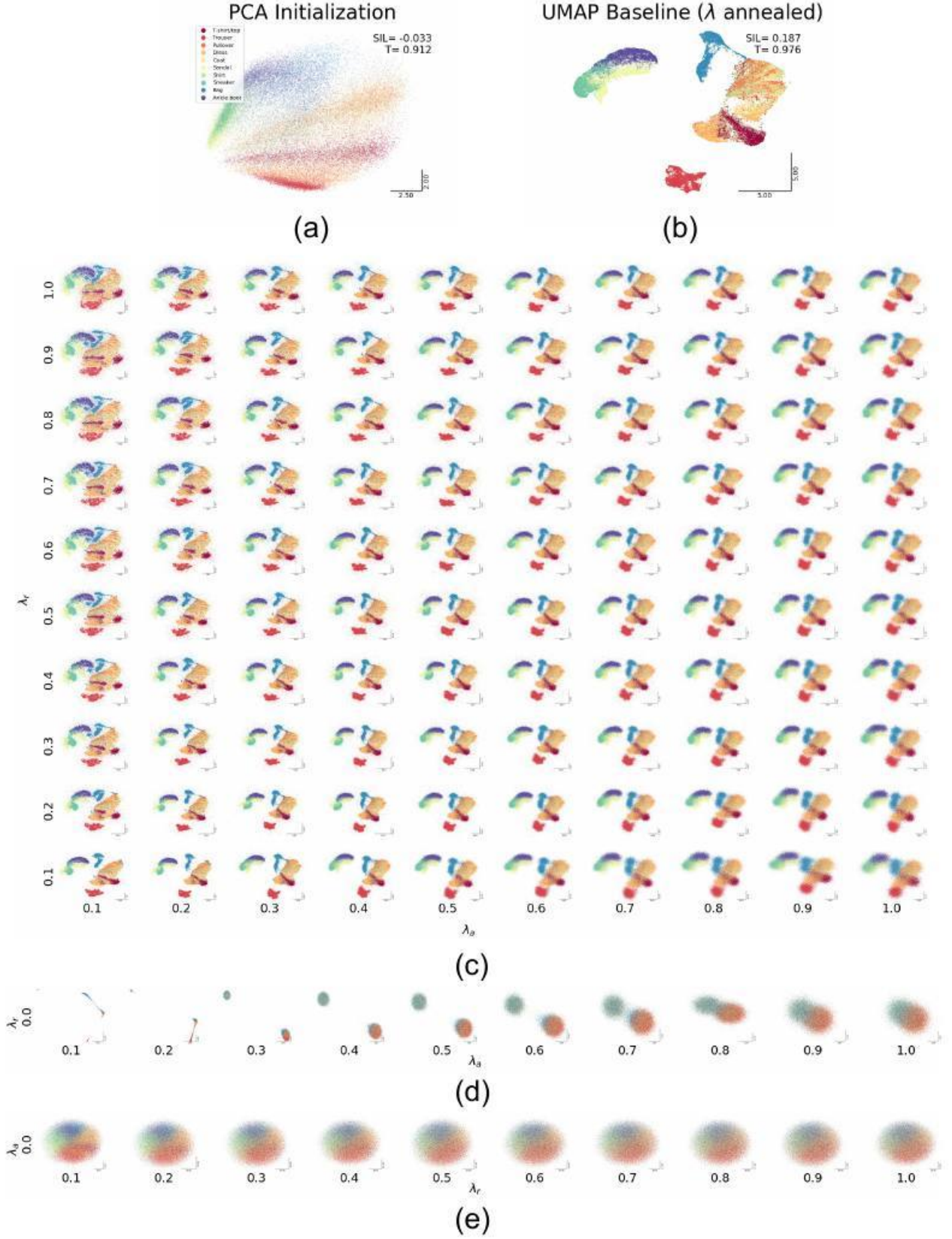


Figure 24: The variation of λ_a and λ_r for the FMNIST dataset using UMAP’s attraction and repulsion shapes. (a) The initialization employed for the embeddings. (b) The baseline obtained when λ is annealed from 1. (c) The embeddings obtained as λ_a and λ_r are varied (without any annealing). (d) When λ_r is set to 0 (no repulsion), the attractive force alone proves unable to produce any cluster. (e) Likewise, when λ_a is set to 0 (no attraction), the repulsive force alone proves unable to produce any clusters.

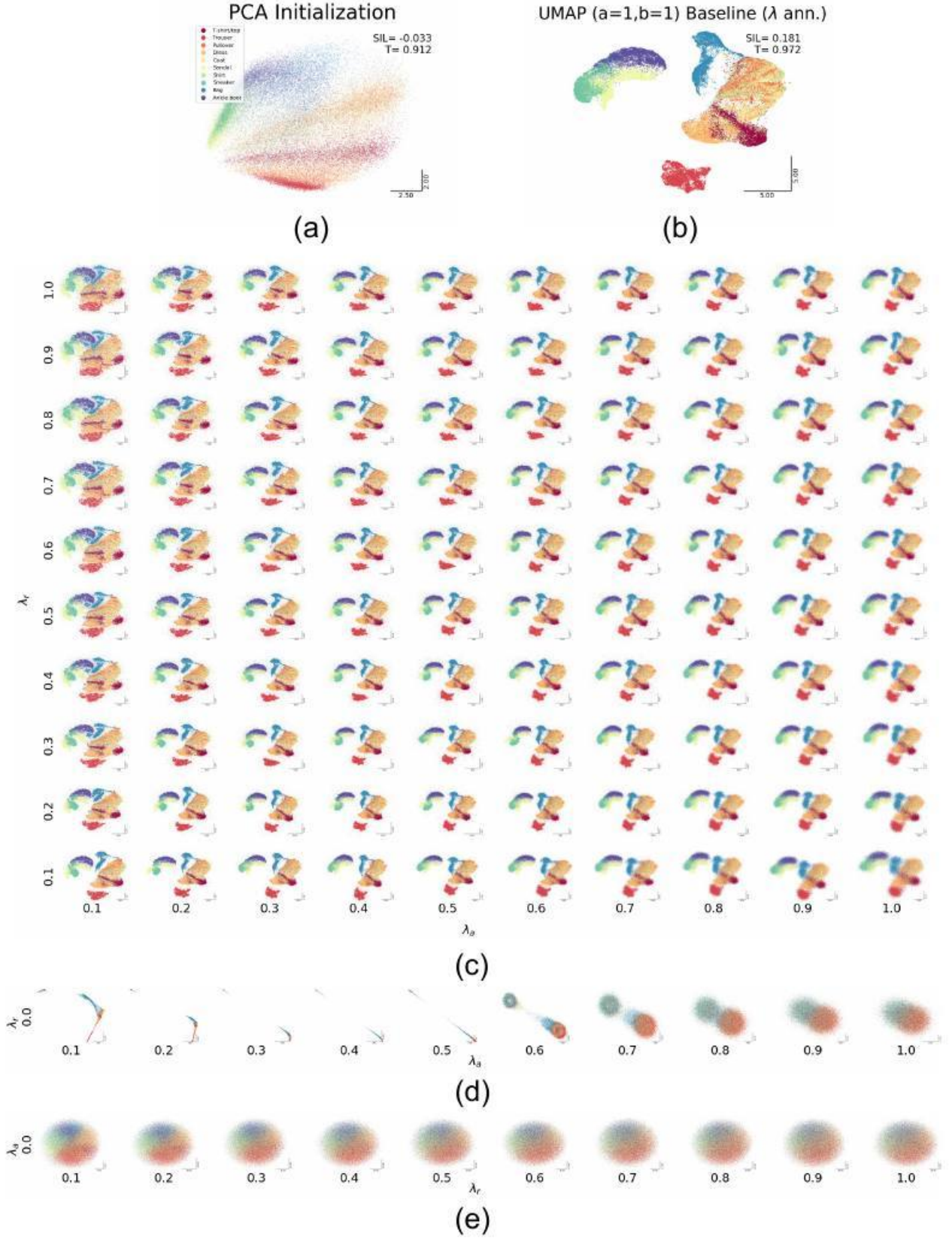


Figure 25: The variation of λ_a and λ_r for the FMNIST dataset using UMAP's attraction and repulsion shapes (by setting $a = 1$ and $b = 1$). (a) The initialization employed for the embeddings. (b) The baseline obtained when λ is annealed from 1. (c) The embeddings obtained as λ_a and λ_r are varied (without any annealing). (d) When λ_r is set to 0 (no repulsion), the attractive force alone proves unable to produce any cluster. (e) Likewise, when λ_a is set to 0 (no attraction), the repulsive force alone proves unable to produce any clusters.

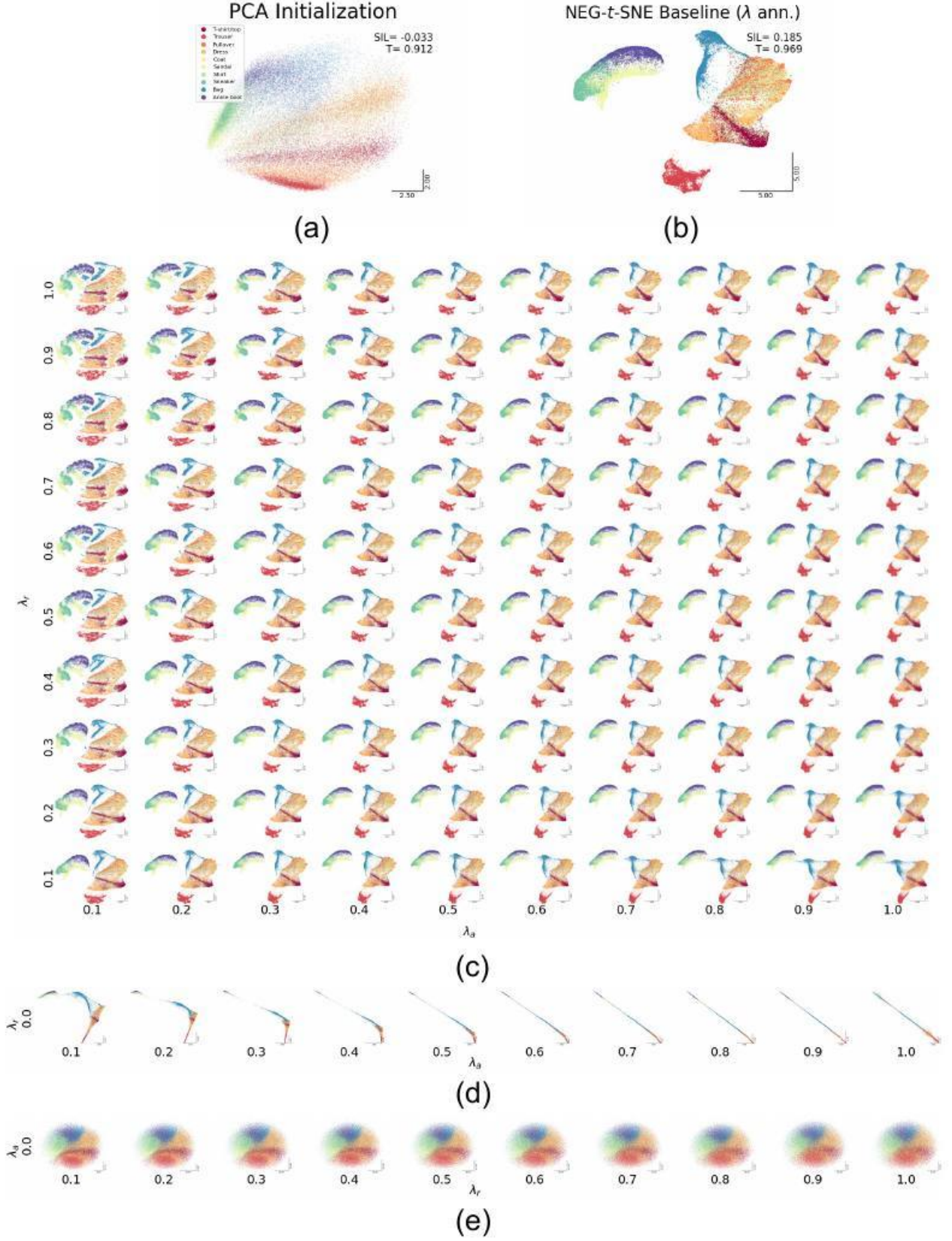


Figure 26: The variation of λ_a and λ_r for the FMNIST dataset using NEG-t-SNE's attraction and repulsion shapes. (a) The initialization employed for the embeddings. (b) The baseline obtained when λ is annealed from 1. (c) The embeddings obtained as λ_a and λ_r are varied (without any annealing). (d) When λ_r is set to 0 (no repulsion), the attractive force alone proves unable to produce any cluster. (e) Likewise, when λ_a is set to 0 (no attraction), the repulsive force alone proves unable to produce any clusters.

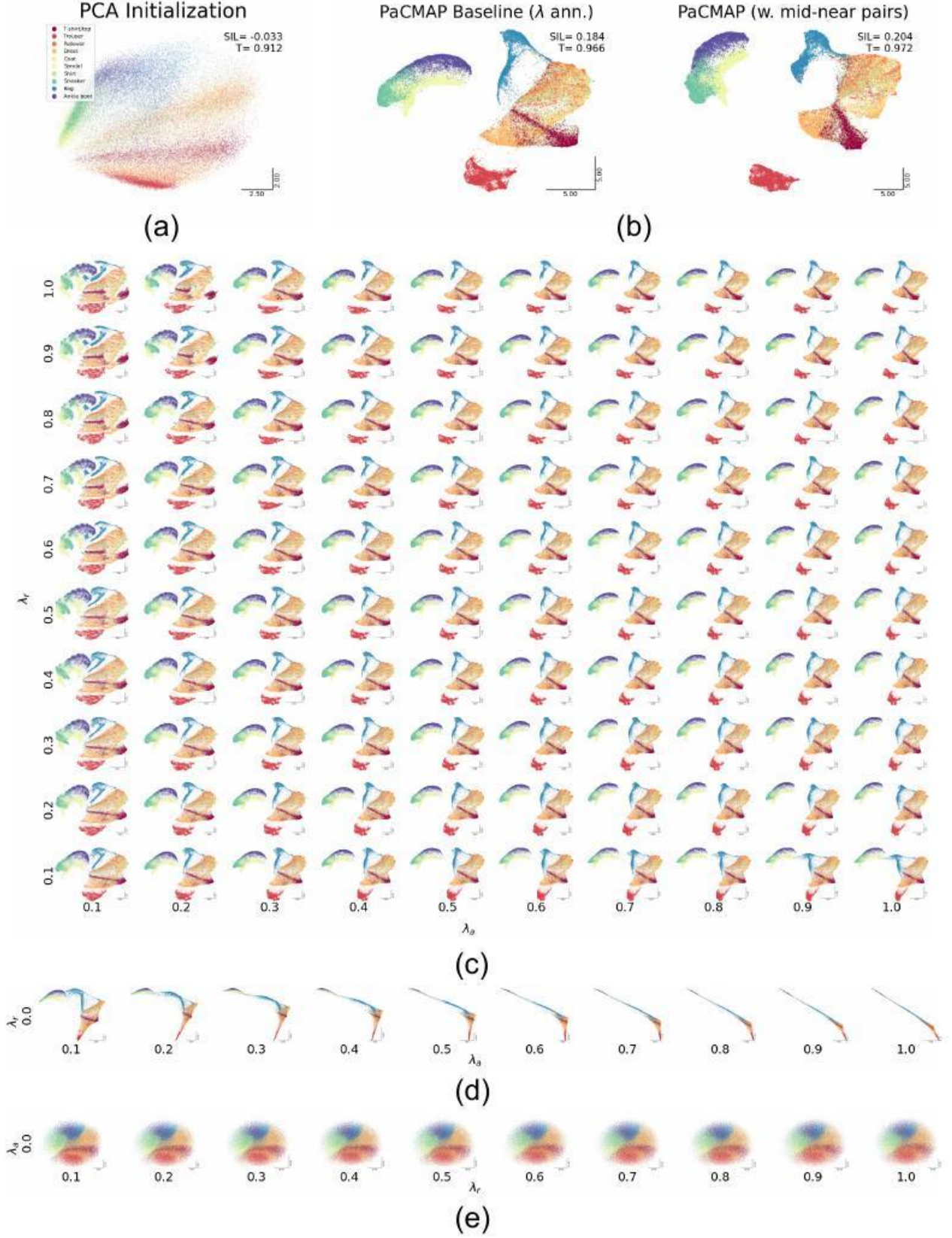


Figure 27: The variation of λ_a and λ_r for the FMNIST dataset using PaCMAP's attraction and repulsion shapes. (a) The initialization employed for the embeddings. (b) Left: the baseline obtained when λ is annealed from 1 (mid-near points being excluded so as to observe the attraction–repulsion interaction in isolation); right: the case in which mid-near points are taken into account. (c) The embeddings obtained as λ_a and λ_r are varied (without any annealing). (d) When λ_r is set to 0 (no repulsion), the attractive force alone proves unable to produce any cluster. (e) Likewise, when λ_a is set to 0 (no attraction), the repulsive force alone proves unable to produce any clusters.

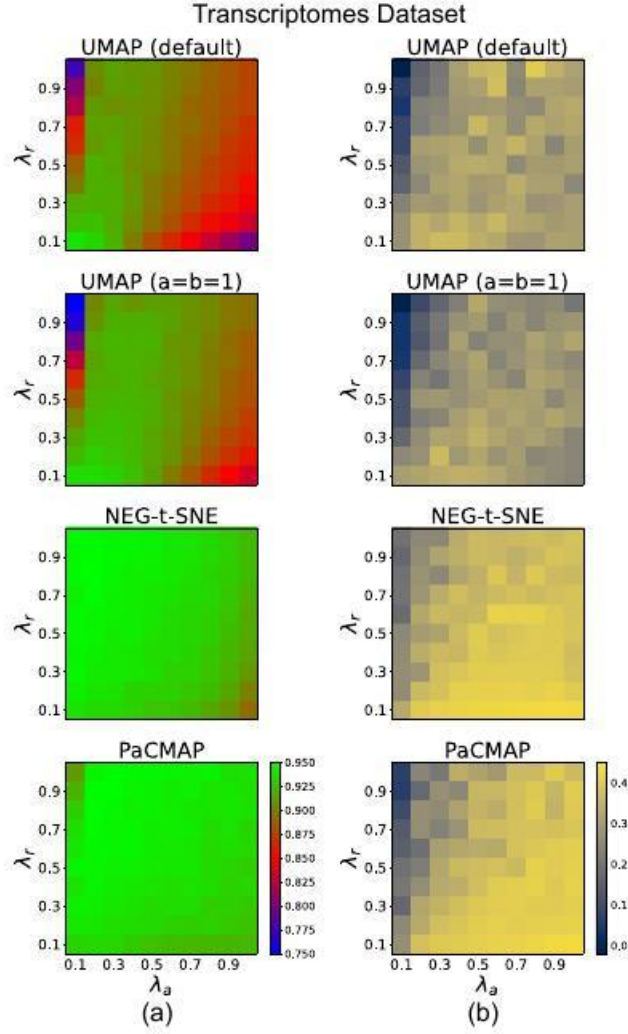


Figure 28: (a) The trustworthiness and (b) the silhouette score attained by the several methods on the single-cell transcriptomes dataset.

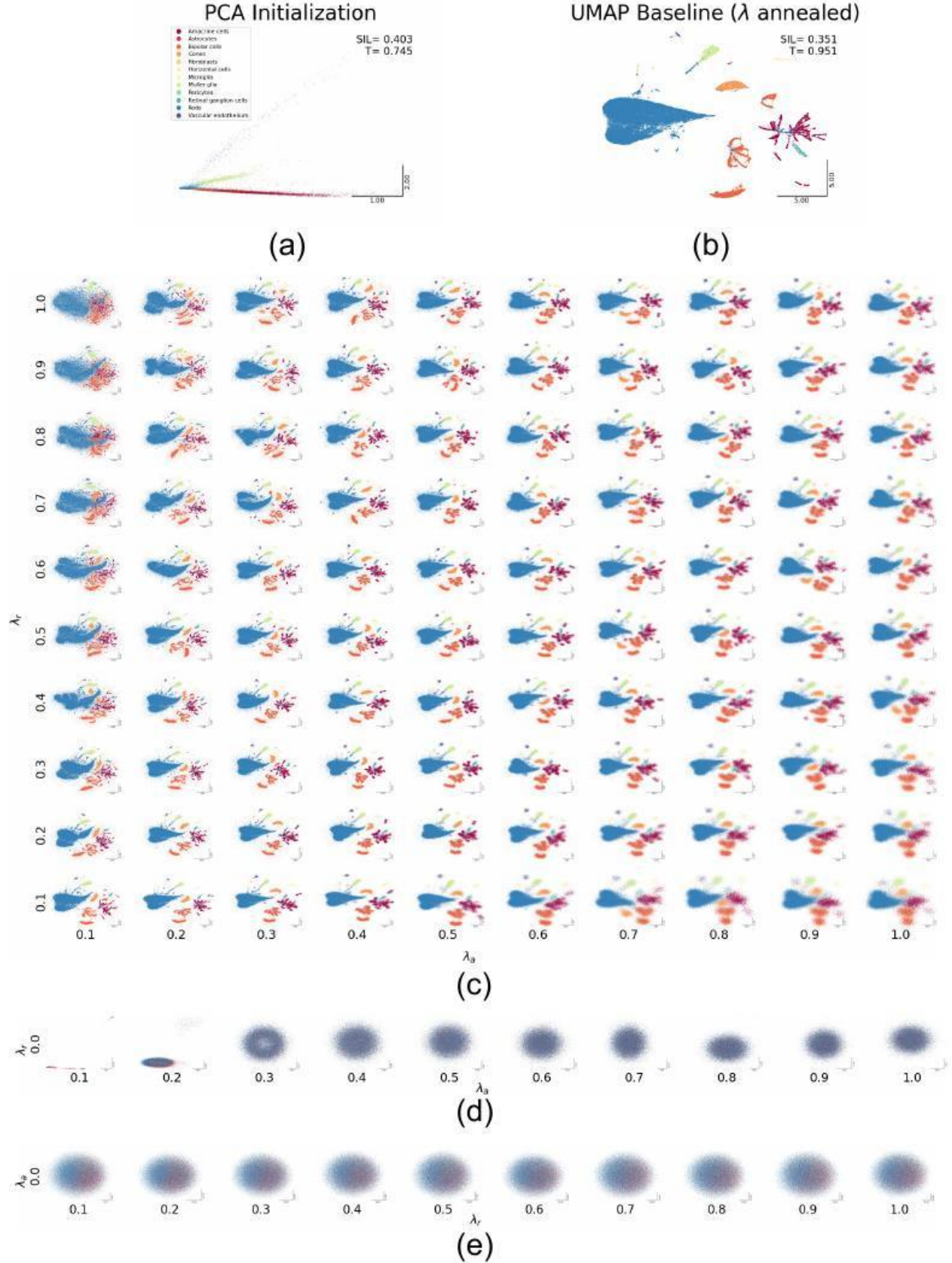


Figure 29: The variation of λ_a and λ_r for the single-cell transcriptomes dataset using UMAP's attraction and repulsion shapes. (a) The initialization employed for the embeddings. (b) The baseline obtained when λ is annealed from 1. (c) The embeddings obtained as λ_a and λ_r are varied (without any annealing). (d) When λ_r is set to 0 (no repulsion), the attractive force alone proves unable to produce any cluster. (e) Likewise, when λ_a is set to 0 (no attraction), the repulsive force alone proves unable to produce any clusters.

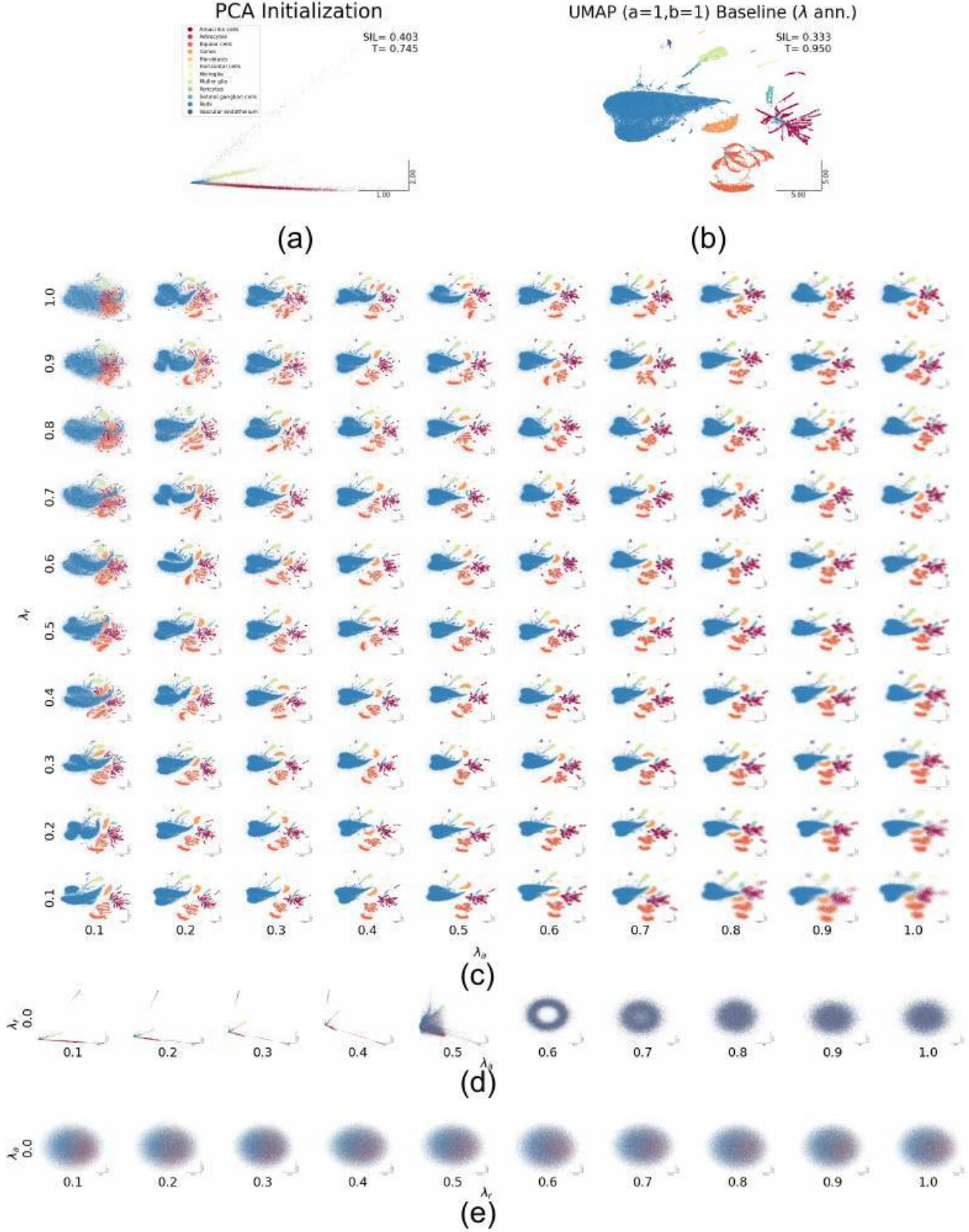


Figure 30: The variation of λ_a and λ_r for the single-cell transcriptomes dataset using UMAP's attraction and repulsion shapes (by setting $a = 1$ and $b = 1$). (a) The initialization employed for the embeddings. (b) The baseline obtained when λ is annealed from 1. (c) The embeddings obtained as λ_a and λ_r are varied (without any annealing). (d) When λ_r is set to 0 (no repulsion), the attractive force alone proves unable to produce any cluster. (e) Likewise, when λ_a is set to 0 (no attraction), the repulsive force alone proves unable to produce any clusters.

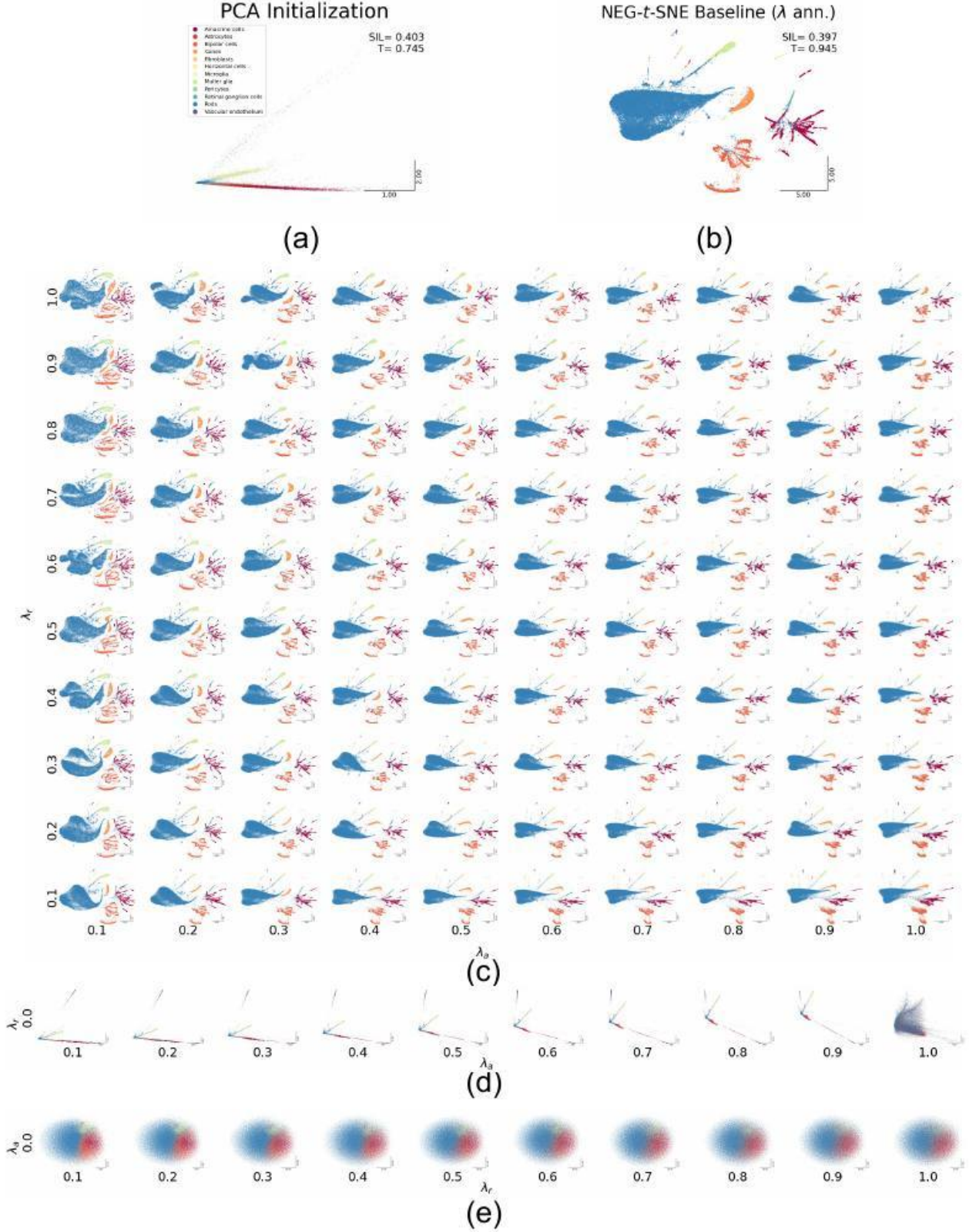


Figure 31: The variation of λ_a and λ_r for the single-cell transcriptomes dataset using NEG-t-SNE's attraction and repulsion shapes. (a) The initialization employed for the embeddings. (b) The baseline obtained when λ is annealed from 1. (c) The embeddings obtained as λ_a and λ_r are varied (without any annealing). (d) When λ_r is set to 0 (no repulsion), the attractive force alone proves unable to produce distinct clusters. (e) Likewise, when λ_a is set to 0 (no attraction), the repulsive force alone proves unable to produce any clusters.

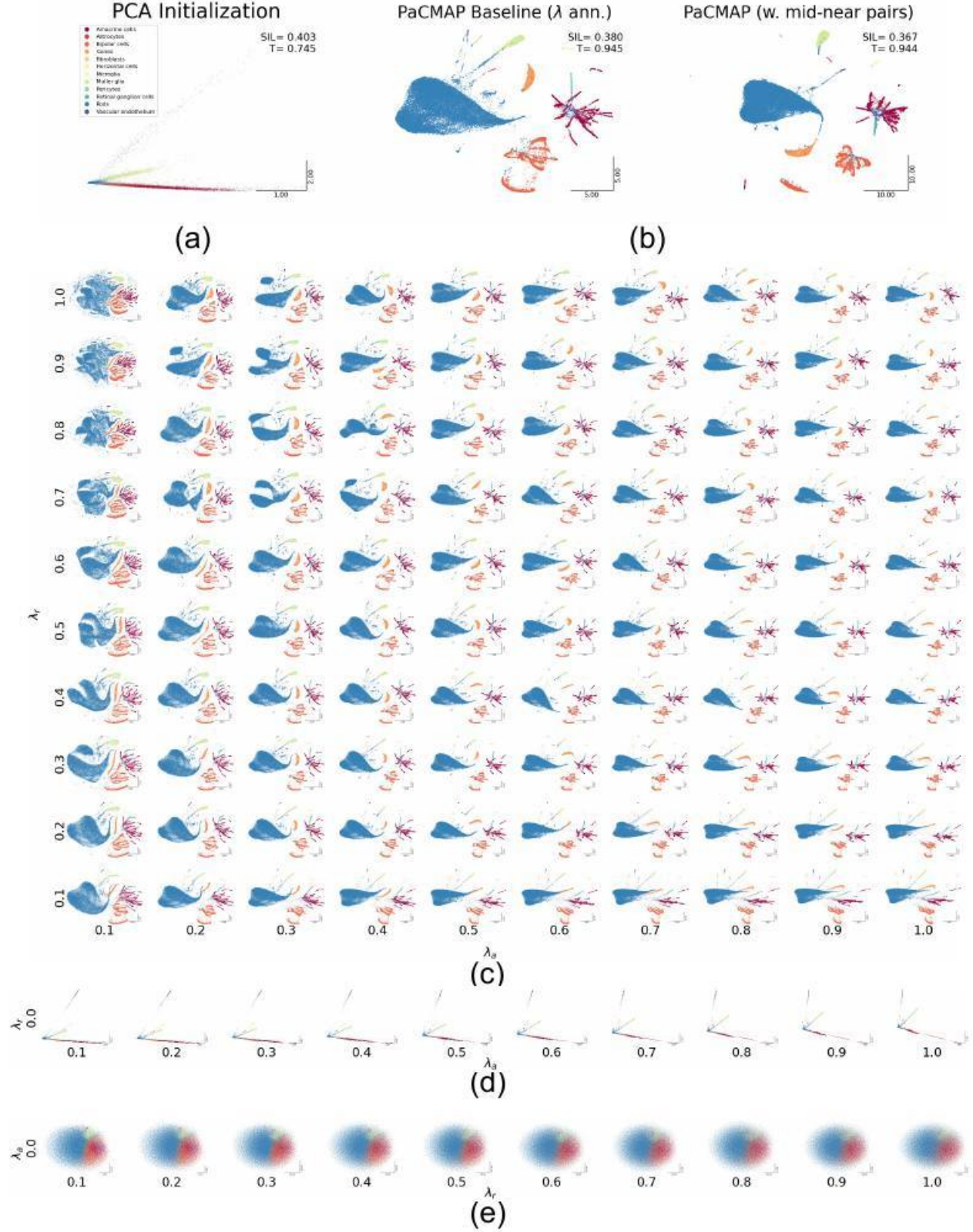


Figure 32: The variation of λ_a and λ_r for the single-cell transcriptomes dataset using PaCMAP's attraction and repulsion shapes. (a) The initialization employed for the embeddings. (b) Left: the baseline obtained when λ is annealed from 1 (mid-near points being excluded so as to observe the attraction–repulsion interaction in isolation); right: the case in which mid-near points are taken into account. (c) The embeddings obtained as λ_a and λ_r are varied (without any annealing). (d) When λ_r is set to 0 (no repulsion), the attractive force alone proves unable to produce any cluster. (e) Likewise, when λ_a is set to 0 (no attraction), the repulsive force alone proves unable to produce any clusters.

J Mix and Match Approach

Throughout our analysis we have regarded the attraction and repulsion shapes as mutually independent—a stance that intimates the possibility of combining attraction and repulsion shapes derived from disparate loss functions and principles. Figure 33 exhibits a handful of such combinations. To this end, we confined the attraction shapes to $[-1, 0]$ (so as to comport with Proposition 4.1), thereby rendering them less sensitive to the learning rate, and we adopted finite repulsion shapes, thus tempering the effect of large repulsive forces at smaller distances. We held the learning rate constant ($\lambda = 1$) and applied the same (UMAP) optimization to all of them, so that any variation might be ascribed to the shapes rather than to learning-rate scheduling or the optimization scheme. On the whole, UMAP’s attraction shape (via $f_a = 0.5f_a^U$) begets superior clusters (a higher silhouette score), whereas PaCMAP’s attraction shape affords superior local structure (a higher trustworthiness)—the latter owing principally to PaCMAP’s attraction shape residing within $[-0.5, 0]$, which averts any flips during the updates. By confining the attraction shapes of UMAP and NEG-*t*-SNE to $[-0.5, 0]$ (via $f_a = 0.25f_a^U$ and $f_a = 0.5f_a^N$, respectively), we may improve the local structure at the expense of clustering performance (Fig. 34).

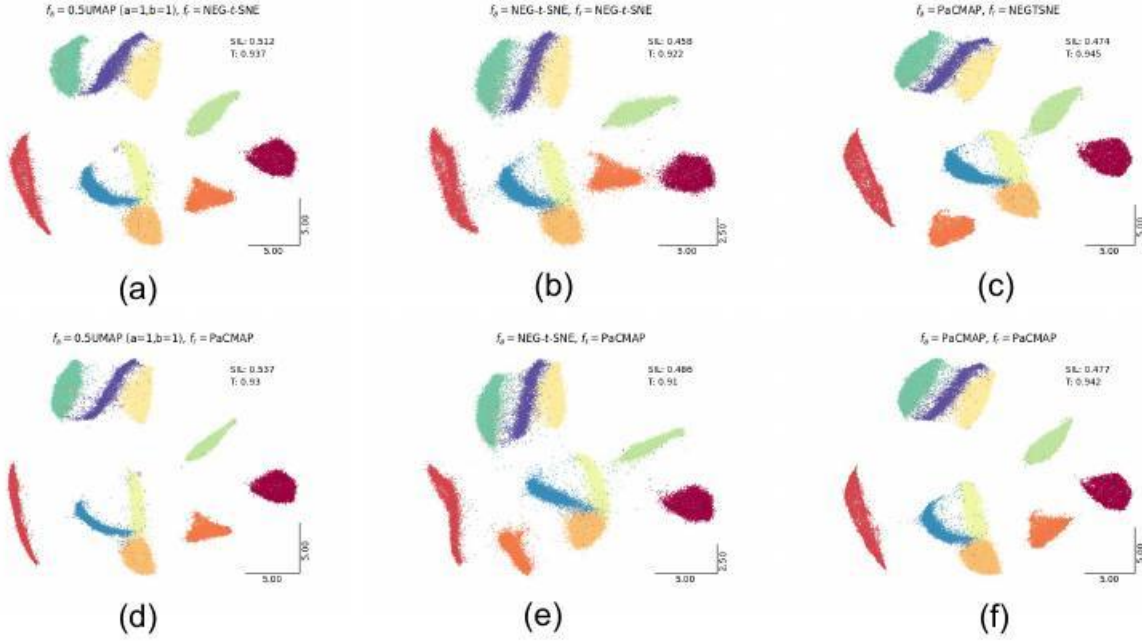


Figure 33: Embedding of MNIST obtained by mixing and matching attraction and repulsion shapes under a constant learning rate ($\lambda = 1$). We employed attraction shapes conforming to Proposition 4.1 together with repulsion shapes that are finite. Top row: the attraction shapes of UMAP, NEG- t -SNE, and PaCMAP paired with the repulsion shape of NEG- t -SNE. Bottom row: the attraction shapes of UMAP, NEG- t -SNE, and PaCMAP paired with the repulsion shape of PaCMAP.

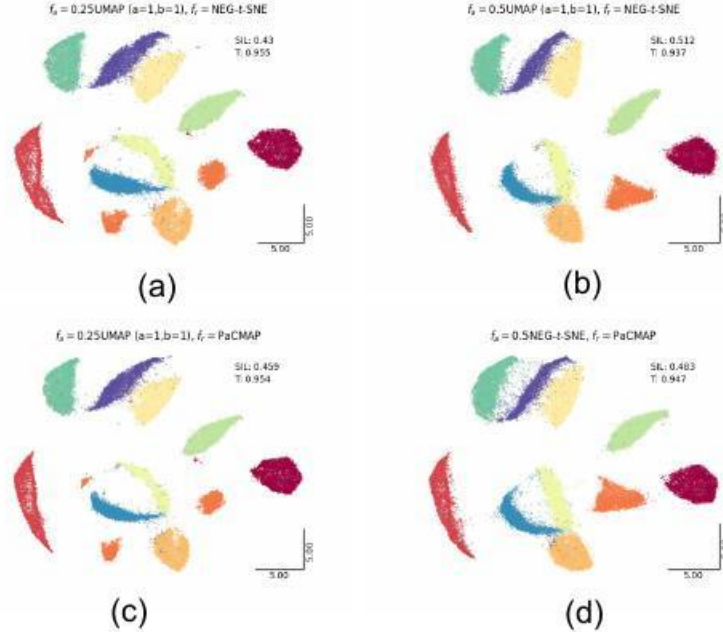


Figure 34: Embedding of MNIST obtained by mixing and matching attraction and repulsion shapes under a constant learning rate ($\lambda = 1$). We confined the attraction shape to $[-0.5, 0]$ so as to preclude any flips during the attractive update. Top row: the attraction shapes of UMAP and NEG- t -SNE paired with the repulsion shape of NEG- t -SNE. Bottom row: the attraction shapes of UMAP and NEG- t -SNE paired with the repulsion shape of PaCMAP.

K Implementation Details

For the purposes of analysis, we implemented our own UMAP algorithm. We availed ourselves of numba (Lam et al. 2015) to compute an exact nearest-neighbor graph (rather than an approximate one) with $k = 15$, together with the scikit-learn (Pedregosa et al. 2011) implementation of the PCA algorithm for PCA initialization. This implementation served to produce and quantify the embeddings in Figs. 1, 2, 3, 4, 6, 8, 9 and 10. We enlisted the same implementation, moreover, when transposing the attraction and repulsion shapes into those of the alternative methods (Figs. 18, 34 unless otherwise stated). The trustworthiness and silhouette scores were computed by means of the corresponding function from the scikit-learn package.

The mappings displayed in Figs. 2, 3, 9, 10 and 15 are rotated onto a reference embedding (panel (a) in each respective figure). To accomplish this, we performed a Procrustes alignment of the embeddings—normalizing them to zero mean and unit norm and then invoking the `orthogonal_procrustes` method of SciPy (Virtanen et al. 2020) to extract the rotation and scaling parameters.

For the comparison with Neg- t -SNE in Fig. 11 we employed the original implementation of the contrastive embedding framework for both the UMAP and the Neg- t -SNE algorithms; the implementation details will be released upon acceptance.

The PaCMAP and LocalMAP embeddings in Figs. 15, 16, 22 (b) (right), 27 (b) (right), and 32 (b) (right) were obtained by means of the official PaCMAP package; the implementation details will be released upon acceptance.

The code for rendering the embeddings of the PubMed dataset in Fig. 7 will be released upon acceptance.